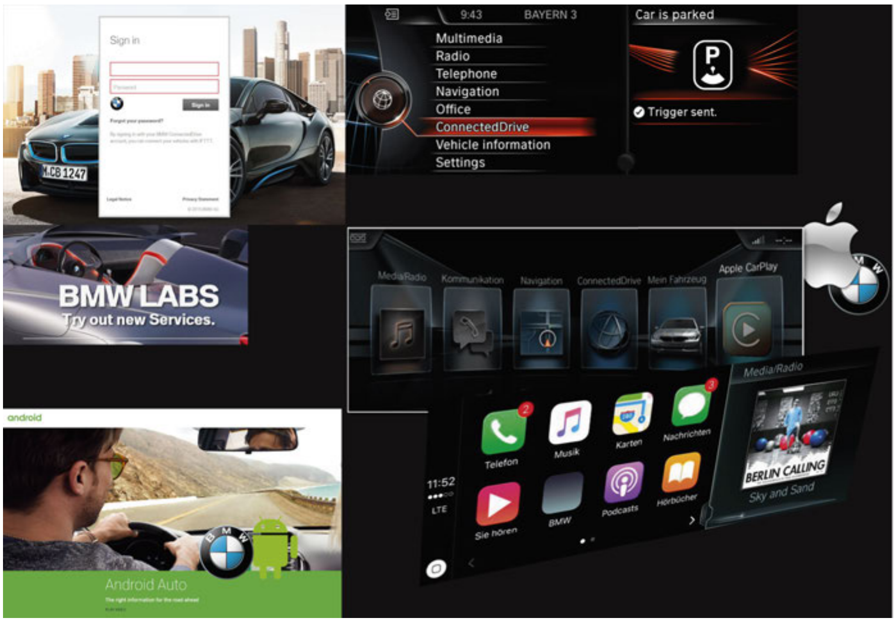
# Technical Training
## Product information.

# Infotainment 2016




Edited for the U.S. market by:
**BMW Group University**
**Technical Training**
ST1610 9/1/2016

**BMW Service**

# General information

### Symbols used

The following symbol is used in this document to facilitate better comprehension or to draw attention to very important information:

⚠

Contains important safety information and information that needs to be observed strictly in order to guarantee the smooth operation of the system.

### Information status and national-market versions

BMW Group vehicles meet the requirements of the highest safety and quality standards. Changes in requirements for environmental protection, customer benefit's and design render necessary continuous development of systems and components. Consequently, there may be discrepancies between the contents of this document and the vehicles available in the training course.

This document basically relates to the European version of left hand drive vehicles. Some operating elements or components are arranged differently in right-hand drive vehicles than shown in the graphics in this document. Further differences may arise as the result of the equipment specification in specific markets or countries.

### Additional sources of information

Further information on the individual topics can be found in the following:

- Owner's Handbook
- Integrated Service Technical Application.

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The information contained in this document forms an integral part of the BMW Group Technical Qualification and is intended for the trainer and participants in the seminar. Refer to the latest relevant information systems of the BMW Group for any changes/additions to the technical data.

Information status: **July 2016**
Technical Training

# Infotainment 2016

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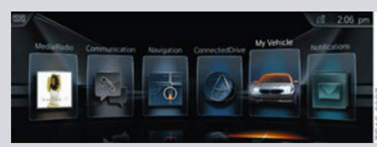
# 1. Variants

## 1.1. User interface of G12

The launch of G12 introduced a significant innovation in the form of a new user interface for the iDrive system at BMW.

Internally, BMW calls this the ID5. To customers it is simply referred to as the **new user interface for the Navigation Professional** system. It’s appearance immediately sets it apart from the list-based predecessor user interfaces.

However, another new user interface (ID6) for the G30 is already planned for 2017.

| Headunit | Interface | Screen mask |
|---|---|---|
| CIC | ID4 |  |
| Headunit High (HU-H) | ID4+ |  |
| HU-H2 (NBT Evo) | ID5 |  |
| HU-H2 (NBT Evo) | ID6 | Anticipated for 2017 |

The ID5 user interface is designed for iDrive control using the controller and for direct touch control by lightly pressing on the Central Information Display (CID).

ID5 will be introduced in numerous additional BMW models throughout 2016.

The so-called RüKo ID5 variants are very similar to the G12 variant.

The touch screen capability of the CID as in G12 is planned for the X5 and X6 (F15, F16) as of August 2016.

### 1.2. Variants with the new user interface

Since March 2016, almost all new BMW models are gradually being equipped with the new user interface.


TE16-0088

User interface ID5

| Model range | E-code | Market introduction | CID touch capability | Apple CarPlay (SA 6CP) |
|---|---|---|---|---|
| 7 Series | G12 | 07/2015 (ID5); * | yes | HU-H2 (SA 609) from 08/16 |
| 6 Series | F06, F06M, F12, F12M, F13, F13M | 03/2016 (ID5); * | no | HU-H2 (SA 609) from 08/16 |
| X3, X4 | F25, F26 | 04/2016 (ID5); * | no | HU-H2 (SA 609) from 08/16 |
| 7 Series PHEV | G12 | 07/2015 (ID5); * | yes | HU-H2 (SA 609) from 08/16 |
| 1 Series and 2 Series (both rear-wheel drive), 3 Series and 4 Series except Z4 | F2x, F3x | 07/2016 (ID5); * | no | HU-H2 (SA 609) from 08/16 |
| 3 Series GT | F34 LCI | 07/2016 (ID5); * | no | HU-H2 (SA 609) from 08/16 |
| X5, X6 | F15, F16 | 08/2016 (ID5); * | yes | HU-H2 (SA 609) from 08/16 |
| X1 | F48 | * | no | HU-H2 (SA 609) ** |
| i3 | I01 | * | no | HU-H2 (SA 609) ** |

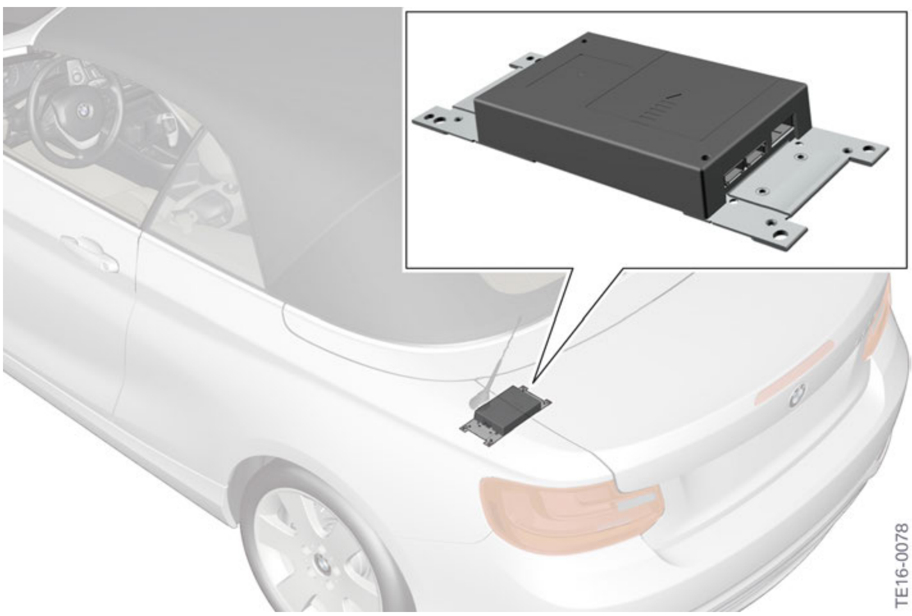
* ID6 in planning for 2017

** Apple CarPlay in planning for 2017

## 1.3. Telematics control unit variants

### 1.3.1. TCB2 luggage compartment variant

In November 2014, BMW introduced a new telematics control unit, the Telematic Communication Box 2, or TCB2 for short. Since it's introduction on the 2 Series convertible (F23), the TCB2 is used as a luggage compartment variant in conjunction with a telematics antenna and a separate emergency GSM antenna (backup) for receiving BMW Teleservices (LTE).




TCB2 as a luggage compartment variant

### 1.3.2. WLAN antenna for the Internet hotspot

A special feature is the hotspot antenna for implementing the Internet hotspot optional equipment in the vehicle. The Internet hotspot was first implemented in October 2015 in the G12 roof variant of TCB2 in combination with the telephone optional equipment.

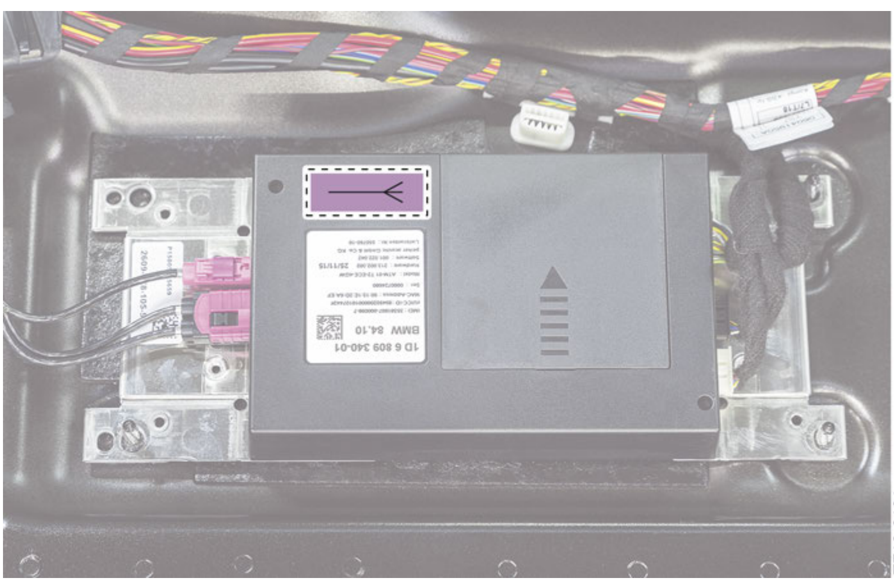
This new technology has also been installed in the 6 Series vehicles as of March 2016. The hotspot antenna, including the Internet hotspot electronics, is now also integrated in the control unit of the luggage compartment variant of the TCB2.

The hotspot antenna and the Internet hotspot fulfil the IEEE standards 802.11 a/b/g and is in the frequency bands of 2.5 GHz and 5 GHz. Up to 10 devices can be connected to the Internet hotspot via WLAN. The Internet hotspot acts as a router (interface to the Internet).The hub or switch functions with which individual end devices could be connected to each other are **deactivated**.

The maximum download rate is 100 Mbit/s.

The Internet hotspot is installed in the vehicle when the customer orders the "Wireless charging" (SA 6NW) or (standard equipment on the 6 Series).

The telematics control unit then contains an integrated WLAN antenna and another electronic component. Together with the two external antennas of the vehicle, these establish the Internet hotspot for access to the World Wide Web.



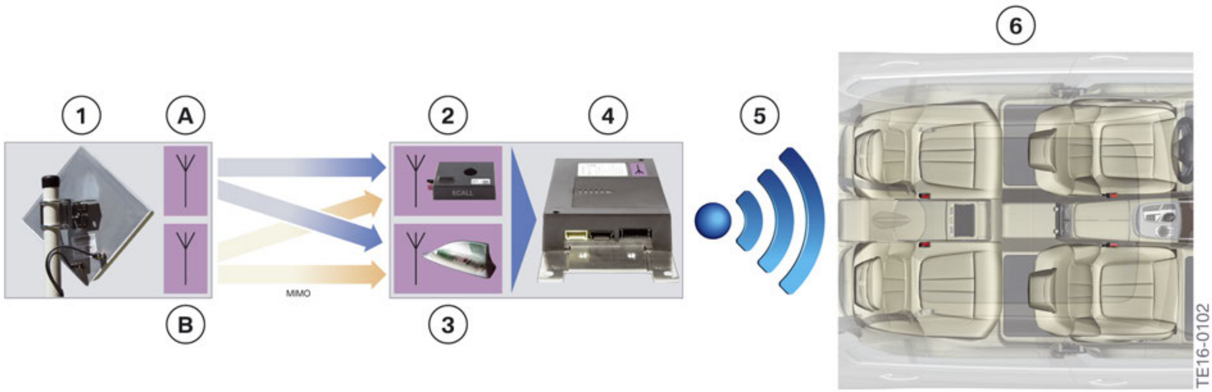
TE16-0101

TCB2 as a luggage compartment variant with the antenna for the Internet hotspot

For data reception via the Internet access in the vehicle, two antennas are required. These are the two telematics antennas connected with the TCB2. The emergency GSM antenna and the shark fin antenna together enable first-rate LTE ( Long Term Evolution) data reception. MIMO technology is used for this.

With the MIMO technology (Multiple Input Multiple Output), it is possible for mobile network providers to offer high data speeds with a low error rate. MIMO is the basis for special coding procedures that not only use the time dimension but also the spatial dimension for information transmission (space-time coding).

As a result, the quality (bit error frequency) and data rate of a wireless connection can be improved significantly. MIMO systems can transmit considerably more bit/s per Hz bandwidth used, and are thus more efficient than conventional systems. In order to realize the MIMO technology at BMW both data antennas present in the vehicle area used simultaneously for the first time for the LTE connection.




MIMO transmit/receive mode via TCB2

| Index | Explanation |
|---|---|
| 1 | LTE transmitter system |
| **A** | LTE antenna 1 |
| **B** | LTE antenna 2 |
| 2 | Emergency GSM antenna |
| 3 | Roof antenna |
| 4 | Telematic Communication Box 2 (TCB2) with MIMO technology |
| 5 | Internet hotspot |
| 6 | Passenger compartment |

### 1.3.3. Provider

AT&T is the data partner provider in the USA for the Internet hotspot of the permanently installed SIM card in the TCB2.

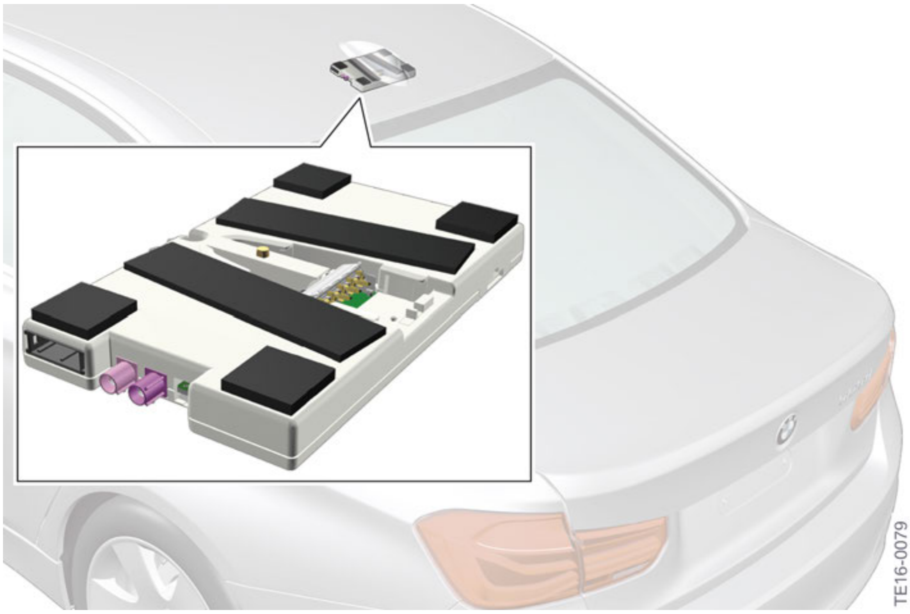
The customer will see this on the new Internet start page when access to the Internet is established via the Internet hotspot in the vehicle. The prices and subscription models are defined by the data partner.

### 1.3.4. TCB2 (roof variant)

In addition to the luggage compartment variant, the TCB2 is also available as a roof variant. The TCB2 roof variant was launched in the BMW 3 Series Saloon (F30 LCI) and in the new 7 Series (G12) in 2015, being positioned as a second variant of the TCB2 in parallel to the luggage compartment variant. This was introduced for package reasons (installation space) and to reduce the number of lines in the vehicle.

# Infotainment 2016

## 1. Variants




Telematic Communication Box 2 (TBC2) as a roof variant

In the roof variant, two telematics antennas integrated in the TCB2 (TEL1 and TEL2) provide LTE data reception. The integrated emergency GSM antenna (backup antenna) also acts as a WLAN antenna for the Internet hotspot.


TCB2 roof variant with integrated hotspot antenna for Internet access in the vehicle

## 1.4. Overview of the various TCB2 variants

The following two tables will explain which variant of the Telematic Communication Box 2 is contained in the individual vehicles.

The table correlates the individual vehicles with the type of system installed – either the **TCB2 luggage compartment variant** or the **TCB2 roof variant.**

### 1.4.1. Overview of the luggage compartment variants

| Model range | E-code | Month of introduction |
|---|---|---|
| 2 Series Convertible | F23 | 11/2014 |
| 2 Series Coupe | F22 | 03/2015 |
| 3 Series Touring, 3 Series Gran Turismo | F31, F34 | 07/2015 |
| 4 Series Coupe, 4 Series Convertible | F32, F33 | 07/2015 |
| 6 Series | F06, F11, F12 | 03/2016 |
| X3 | F25 | 04/2016 |
| BMWi i3 | I01 | planned for 2017 |

### 1.4.2. Overview of the roof variants

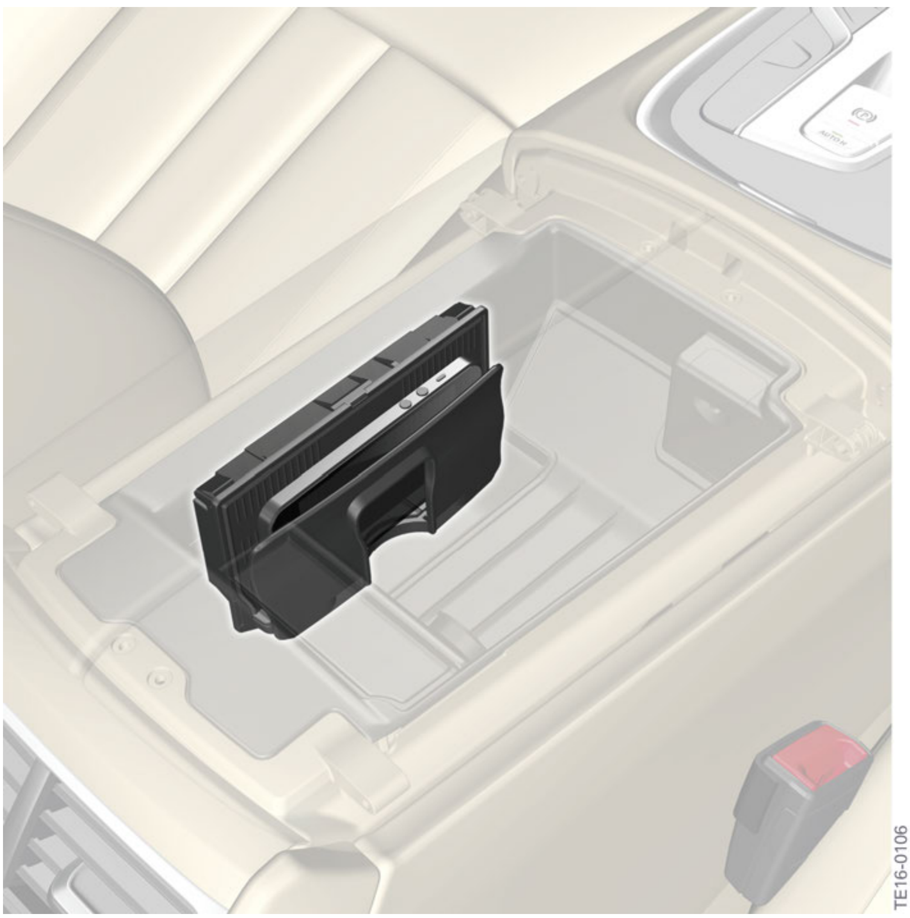
| Model range | E-code | Month of introduction |
|---|---|---|
| 3-Series Saloon | F30 LCI | 07/2015 |
| X1 | F48 | 07/2015 |
| 4 Series Gran Coupé | F36 | 07/2015 |
| 7 Series | G12 | 10/2015 |
| X4 | F26 | 04/2016 |
| X5, X6 | F15, F16 | 08/2016 |

## 1.5. Wireless charging

### 1.5.1. Use with telephone optional equipment

The previously mentioned Internet hotspot is always a component of one of the two telephone optional equipment packages: either the "Hands-free Bluetooth and USB audio connection" (SA 6NH) or "Wireless Charging" (SA 6NW).

In addition, in the US market, the "Wireless charging" optional equipment (SA 6NW) fully replaces the "Convenient telephony" optional equipment (snap-an adapter).




Wireless charging station in the G12

"Telephone with wireless charging" (WCA) consists of a wireless charging station and a line compensator to improve mobile network reception. This combination has been available in BMW vehicles since G12.

The mobile phone of the customer can be inductively charged in the vehicle if the phone is equipped with this capability. Moreover, a wireless connection is additionally established between the mobile phone and exterior antenna. To improve reception quality and compensate for charging losses by the wireless charging station, the vehicle is additionally equipped with a line compensator.

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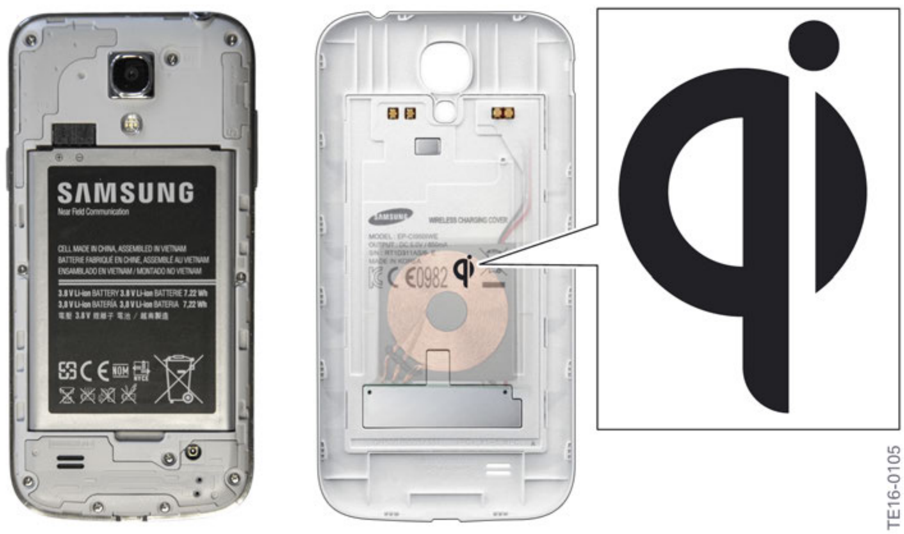
## 1. Variants

The wireless charging feature can only be used with mobile phones with the Qi® standard. This standard can be identified by the Qi® symbol on the rear side or on the additional equipment (charging cover) of the mobile phone/accessory in question.

This standard ensures that the chargers are compatible with the wireless charging station in the vehicle




Qi symbol inside the rear cover of the Samsung S4

For more information on the compatibility of mobile phones, see section 1.6.3 of this product information.

The BMW 6 Series in March 2016 and the X3 and X4 in April 2016 extended the spectrum of vehicles containing a wireless charging station, or WCA (SA 6NW). This technology is installed here in the same manner as the wireless charging station in the BMW 7 Series (G12). A protective cover with integrated WCA technology has been developed for the iPhone® from Apple® by BMW Parts and Accessories. Further details on the new wireless charging iPhone cover are available on Shopbmwusa.com.

### 1.5.2. Wireless charging station design update

#### BMW

A special version of the wireless charging station (WCA) was designed specifically for the **BMW 1 Series, 2 Series, 3 Series and 4 Series** (F3x, F8x) due to the available space on these vehicles; this version is referred to in the development department as the Rüko variant (Rüko = backward compatible).

It has been available to all vehicles specified above as of July 2016.


Wireless charging station for BMW 1 Series, 2 Series, 3 Series and 4 Series

| Index | Explanation |
|---|---|
| 1 | Wireless charging station for BMW 1 Series, 2 Series, 3 Series and 4 Series as from July 2016 |
| 2 | Line compensator in the rear area of the vehicle |

**Note:**

An exception is the BMW X1 from the UKL2 (F48). These vehicles will get a specially modified version of the wireless charging station (WCA) next year.

### 1.5.3. WCA structure and vehicle electrical system connection

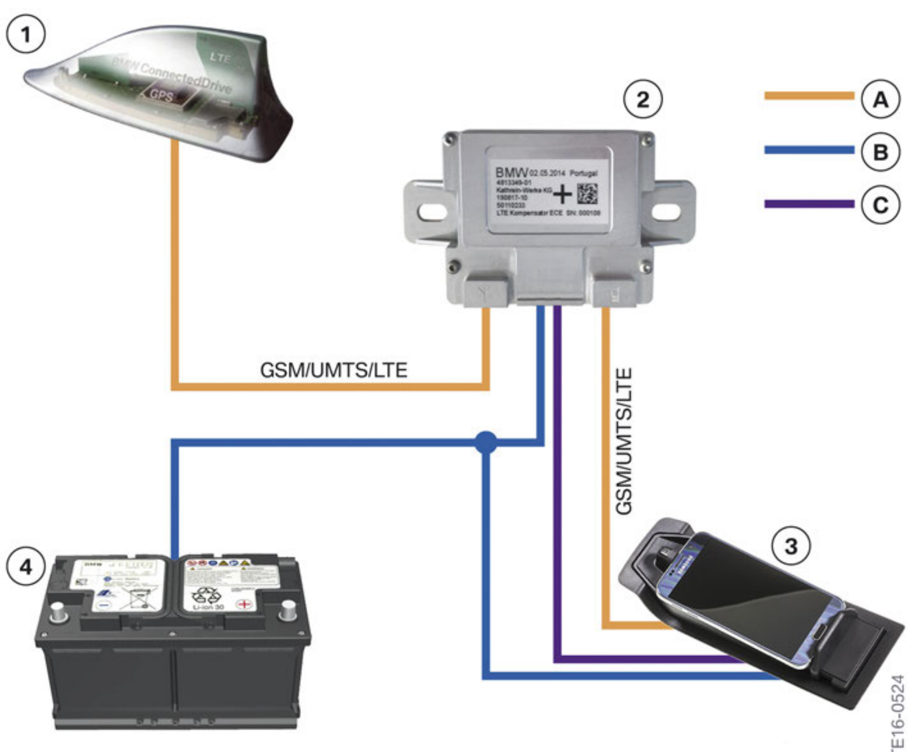
A spring mechanism clamps the mobile phone into the wireless charging station. The electrical connections of the WCA are shown the following graphic.


Wireless charging station for BMW 1 Series, 2 Series, 3 Series and 4 Series with the electrical connections

| Index | Explanation |
|---|---|
| 1 | Voltage supply and activation line |
| 2 | Antenna connection to the line compensator and on to the external antenna |

Using the previously shown plug connections, the wireless charging station is connected to the vehicle electrical system at multiple locations.

The newly developed WCA is incorporated as follows:



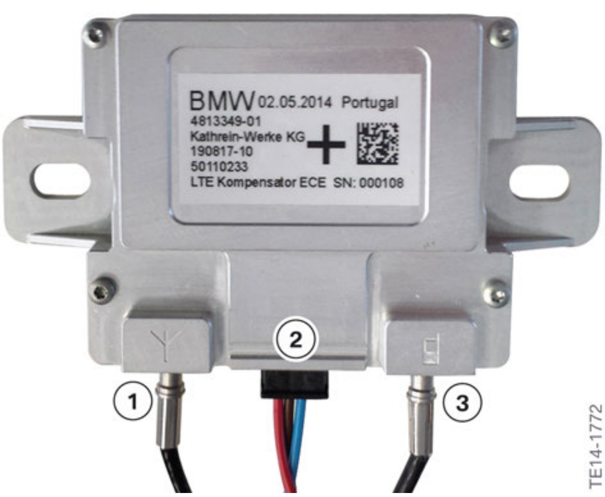

Connection of the wireless charging station to the vehicle electrical system

| Index | Explanation |
|---|---|
| A (yellow) | Antenna lines |
| B (blue) | Voltage supply |
| C (violet) | Activation line |
| 1 | Roof antenna |
| 2 | Line compensator |
| 3 | Wireless charging station with inserted mobile phone |
| 4 | Vehicle battery |

### 1.5.4. Line compensator

An LTE-capable line compensator is installed to compensate for the losses due to wireless charging. With this line compensator, the reception quality of the wireless charging transmit/receive system again reaches the level of a conventional phone with a snap-in adapter.

In vehicles of the BMW 1 Series, 2 Series, 3 Series and 4 Series with the WCA optional equipment, the line compensator is installed at various locations in the luggage compartment. For details, please see the repair instructions in the BMW diagnosis system.




Line compensator with vehicle electrical system connection

| Index | Explanation |
|---|---|
| 1 | Antenna port to mobile network antenna in the roof antenna |
| 2 | Line compensation voltage supply, activation line |
| 3 | Antenna port for wireless charging station |

### 1.5.5. Changed LED logic

Unlike the already familiar WCA station (wireless charging station in the center console in G12, 6 Series and X3, X4), the WCA Rüko station presented here uses a different flashing/color LED logic.

The LED always lights up blue for **ten seconds** at the start due to the self-diagnosis routine. However, the slide must be held for this period or an object must be inserted. When the slide is release, the LED goes out.

If an object is inserted in the wireless charging station and if this object is **not** a mobile phone that meets the Qi® standard, the LED goes out and turns black.

If everything is OK, the charging process is started and the **LED lights up blue**.

If the **LED lights up orange**, this indicates overheating of the mobile phone or the presence of a foreign body.

⚠

If a Qi®-compatible device and another metallic object (e.g. a coin) are located on the wireless charging station, the charging process is aborted after a short period.

If connection to the exterior antenna is requested via the capacitive antenna coupling with the mobile phone, the exterior antenna line and the line compensator are monitored additionally.

If the **LED lights up red** instead of blue, the antenna line is not connected. The red LED then lights up for 23 seconds. If customers ask about this, it is communicated to them as a "internal error". In the Owner's Manual, the customer is asked to contact the next BMW service workshop to have the fault corrected.

⚠

The Qi® standard is not fully identical in all devices. Currently, the new wireless charging station in combination with current Android® mobile phones from various manufacturers can result in problems with the charging procedure.

In G12, the BMW 6 Series and X3 and X4 with the center console tray, a different wireless charging coil technology is used in which these problems do not occur.

For all BMW wireless charging stations, it is important that the Qi® standard is fulfilled 100%. Thus, the problem is not caused by the vehicle but by the mobile phone.

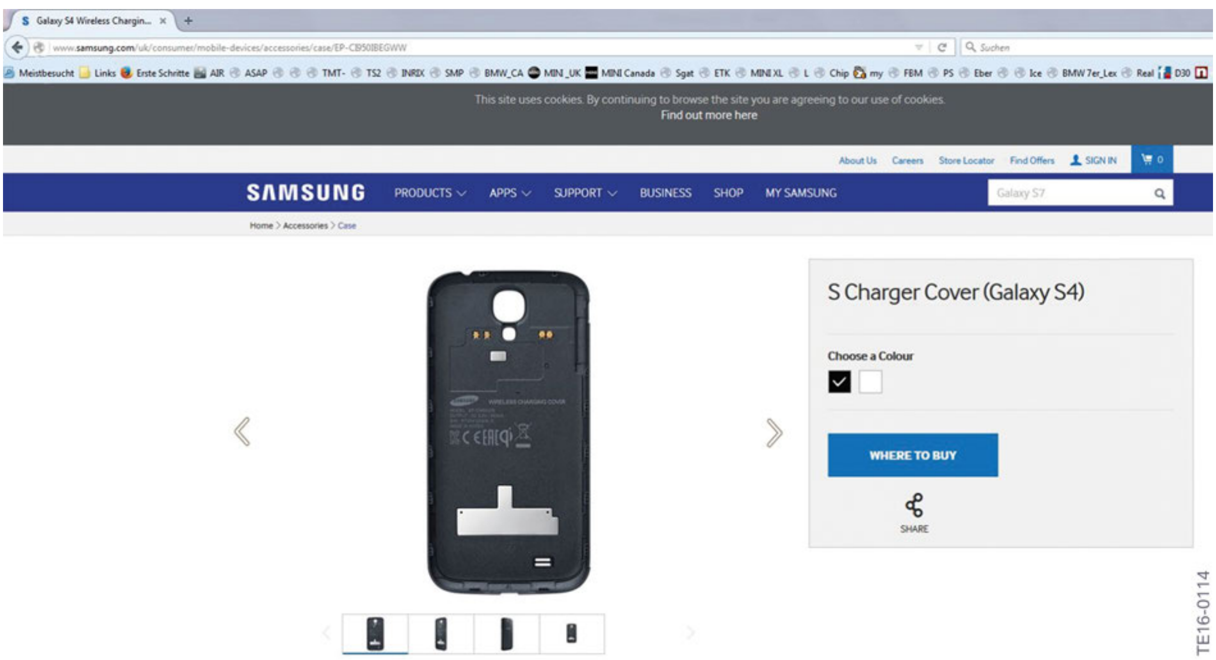
## 1.6. Smartphones capable of wireless charging

While Android devices have been capable of wireless charging since 2013, wireless charging with the Qi standard is neither used nor is being prepared for in the Apple iPhone. Several manufacturers of Android® devices have already made the necessary preparations in their devices.

### 1.6.1. Android devices

Many Android smartphones are already equipped with the necessary contacts in the back of the phone for wireless charging. Suitable rear covers for WCA are supplied in the original accessories (OEM).

At Samsung, for example, covers are offered directly on the manufacturer's website to make the most recent models capable of wireless charging. The most recent devices are already capable of wireless charging as a standard feature and in other cases, just the cover with the receiver coil is needed to use the function.




TE16-0114

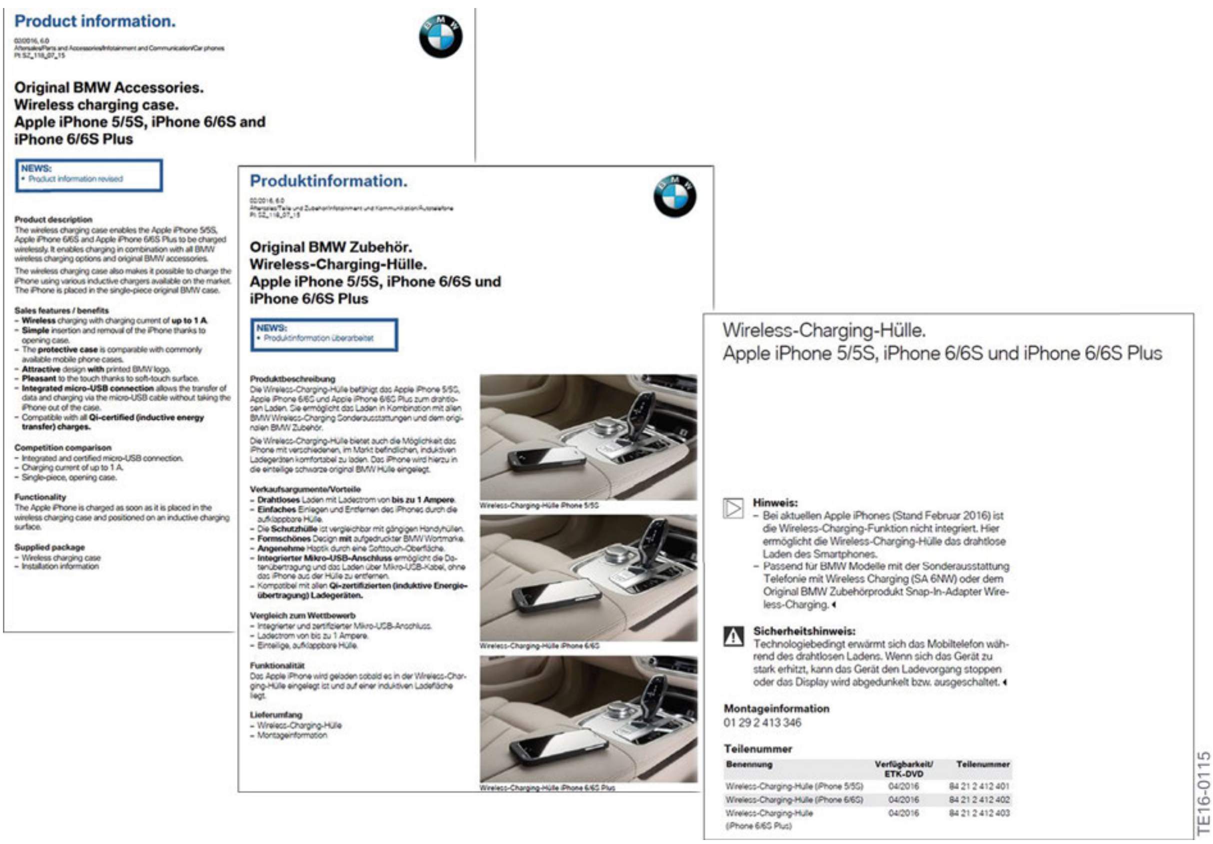
Example of the wireless charging cover for the Samsung S4 (source: Samsung homepage)

### 1.6.2. Apple devices

To be able to use the wireless charging function with an Apple iPhone beginning with iPhone 5, BMW offers it's customers the option of using a wireless charging adapter from it's parts and accessories.

The iPhone is inserted into this wireless charging case as would be into a protective case, and can then be charged in a wireless manner.

For details, see the PI product information: SZ_118_07_15.



Product information.

Original BMW Accessories.
Wireless charging case.
Apple iPhone 5/5S, iPhone 6/6S and iPhone 6/6S Plus

NEWS:
- Product information revised

**Product description**

The wireless charging case enables the Apple iPhone 5/5S, Apple iPhone 6/6S and Apple iPhone 6/6S Plus to be charged wirelessly. It enables charging in combination with all BMW wireless charging options and original BMW accessories.

The wireless charging case also makes it possible to charge the iPhone using various inductive chargers available on the market. The iPhone is placed in the single-piece original BMW case.

**Sales features / benefits**
- **Wireless** charging with charging current of **up to 1 A**.
- **Simple** insertion and removal of the iPhone thanks to opening case.
- The **protective case** is comparable with commonly available mobile phone cases.
- **Attractive** design **with** printed BMW logo.
- **Pleasant** to the touch thanks to soft-touch surface.
- **Integrated micro-USB connection** allows the transfer of data and charging via the micro-USB cable without taking the iPhone out of the case.
- Compatible with all **Qi-certified (inductive energy transfer) charges.**

**Competition comparison**
- Integrated and certified micro-USB connection.
- Charging current of up to 1 A.
- Single-piece, opening case.

**Functionality**

The Apple iPhone is charged as soon as it is placed in the wireless charging case and positioned on an inductive charging surface.

**Supplied package**
- Wireless charging case
- Installation information

Produktinformation.

Original BMW Zubehör.
Wireless-Charging-Hülle.
Apple iPhone 5/5S, iPhone 6/6S und iPhone 6/6S Plus

NEWS:
- Produktinformation überarbeitet

**Produktbeschreibung**

Die Wireless-Charging-Hülle befähigt das Apple iPhone 5/5S, Apple iPhone 6/6S und Apple iPhone 6/6S Plus zum drahtlosen Laden. Sie ermöglicht das Laden in Kombination mit allen BMW Wireless-Charging Sonderausstattungen und dem originalen BMW Zubehör.

Die Wireless-Charging-Hülle bietet auch die Möglichkeit das iPhone mit verschiedenen, im Markt befindlichen, induktiven Ladegeräten komfortabel zu laden. Das iPhone wird hierzu in die einteilige schwarze original BMW Hülle eingelegt.

**Verkaufsargumente/Vorteile**
- **Drahtloses** Laden mit Ladestrom von **bis zu 1 Ampere**.
- **Einfaches** Einlegen und Entfernen des iPhones durch die aufklappbare Hülle.
- Die **Schutzhülle** ist vergleichbar mit gängigen Handyhüllen.
- **Formschönes** Design **mit** aufgedruckter BMW Wortmarke.
- **Angenehme** Haptik durch eine Softtouch-Oberfläche.
- **Integrierter Mikro-USB-Anschluss** ermöglicht die Datenübertragung und das Laden über Mikro-USB-Kabel, ohne das iPhone aus der Hülle zu entfernen.
- Kompatibel mit allen **Qi-zertifizierten (induktive Energieübertragung) Ladegeräten.**

**Vergleich zum Wettbewerb**
- Integrierter und zertifizierter Mikro-USB-Anschluss.
- Ladestrom von bis zu 1 Ampere.
- Einteilige, aufklappbare Hülle.

**Funktionalität**

Das Apple iPhone wird geladen sobald es in der Wireless-Charging-Hülle eingelegt ist und auf einer induktiven Ladefläche liegt.

**Lieferumfang**
- Wireless-Charging-Hülle
- Montageinformation

Wireless-Charging-Hülle iPhone 5/5S

Wireless-Charging-Hülle iPhone 6/6S

Wireless-Charging-Hülle iPhone 6/6S Plus

Wireless-Charging-Hülle.
Apple iPhone 5/5S, iPhone 6/6S und iPhone 6/6S Plus

**Hinweis:**
- Bei aktuellen Apple iPhones (Stand Februar 2016) ist die Wireless-Charging-Funktion nicht integriert. Hier ermöglicht die Wireless-Charging-Hülle das drahtlose Laden des Smartphones.
- Passend für BMW Modelle mit der Sonderausstattung Telefonie mit Wireless Charging (SA 6NW) oder dem Original BMW Zubehörprodukt Snap-In-Adapter Wireless-Charging.

**Sicherheitshinweis:**
Technologiebedingt erwärmt sich das Mobiltelefon während des drahtlosen Ladens. Wenn sich das Gerät zu stark erhitzt, kann das Gerät den Ladevorgang stoppen oder das Display wird abgedunkelt bzw. ausgeschaltet.

**Montageinformation**
01 29 2 413 346

**Teilenummer**

| Benennung | Verfügbarkeit/ ETK-DVD | Teilenummer |
|---|---|---|
| Wireless-Charging-Hülle (iPhone 5/5S) | 04/2016 | 84 21 2 412 401 |
| Wireless-Charging-Hülle (iPhone 6/6S) | 04/2016 | 84 21 2 412 402 |
| Wireless-Charging-Hülle (iPhone 6/6S Plus) | 04/2016 | 84 21 2 412 403 |

TE16-0115

Original BMW accessories product information for the wireless charging case

A protective cover with integrated WCA technology has been developed for the iPhone® from Apple® by BMW Parts and Accessories. Further details on the new wireless charging iPhone cover are available on Shopbmwusa.com.

Shopbmwusa.com

# Infotainment 2016

## 1. Variants


BMW wireless charging cell phone case for iPhone available on Shopbmwusa.com

### 1.6.3. Compatible devices

There is again a reference guide for customers so that they can find out whether their smartphone is compatible with the G12.

The BMW My-CE page at:

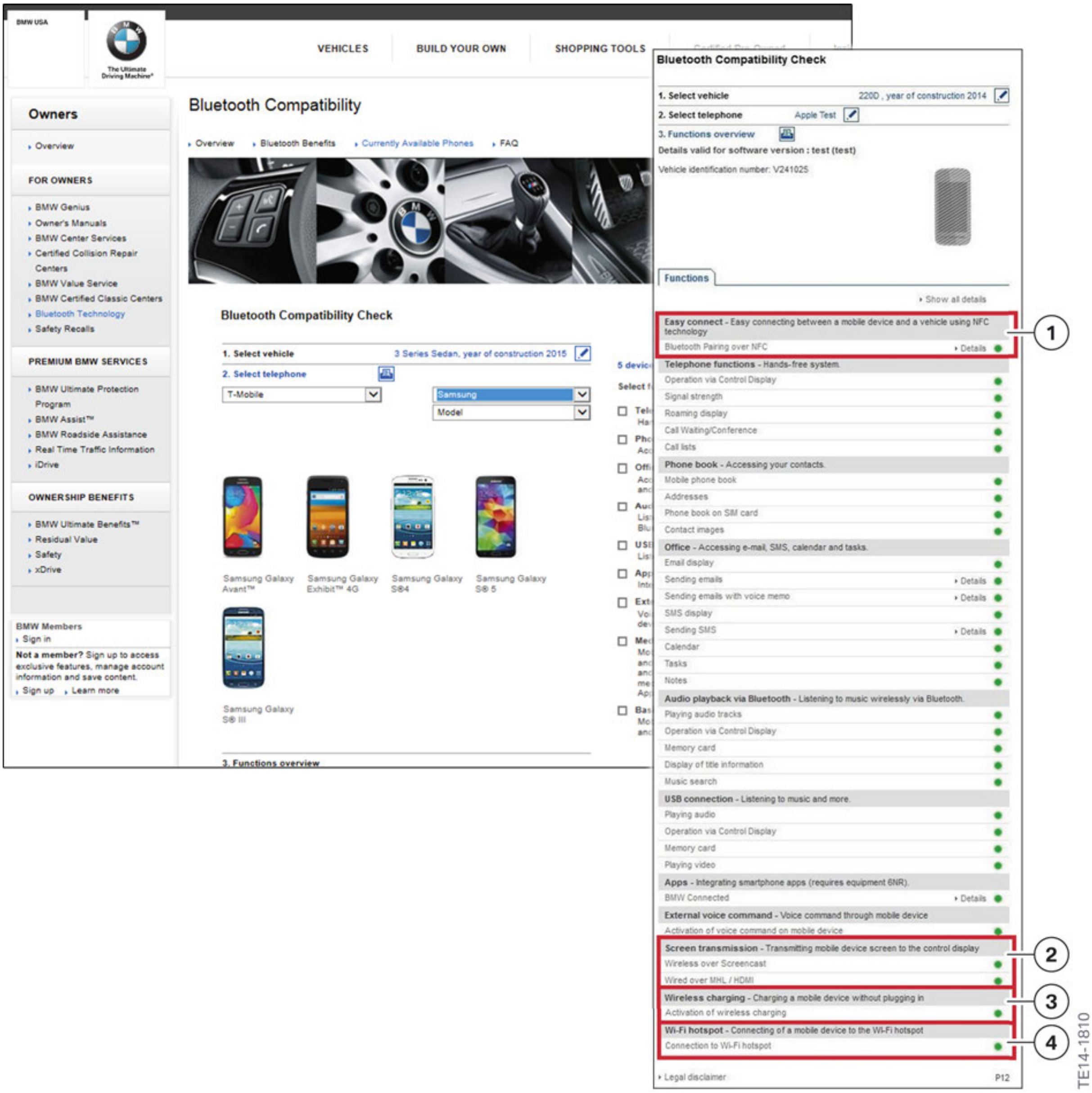
http://www.bmwusa.com/Standard/Content/Owner/BluetoothTechnology/bluetoothframedin.aspx

www.bmw.com/bluetooth

The web site provides information about **Wireless Charging (WCA)** in addition to the already familiar compatibility information. New here is the information about compatibility for Wi-Fi Hotspot, Near Field Communication (NFC) and Screencast.

# Infotainment 2016

## 1. Variants




MY-CE information page for the customer

| Index | Explanation |
|---|---|
| 1 | Near Field Communication (NFC) |
| 2 | Screencast |
| 3 | Wireless charging |
| 4 | WLAN hotspot |

## Infotainment 2016

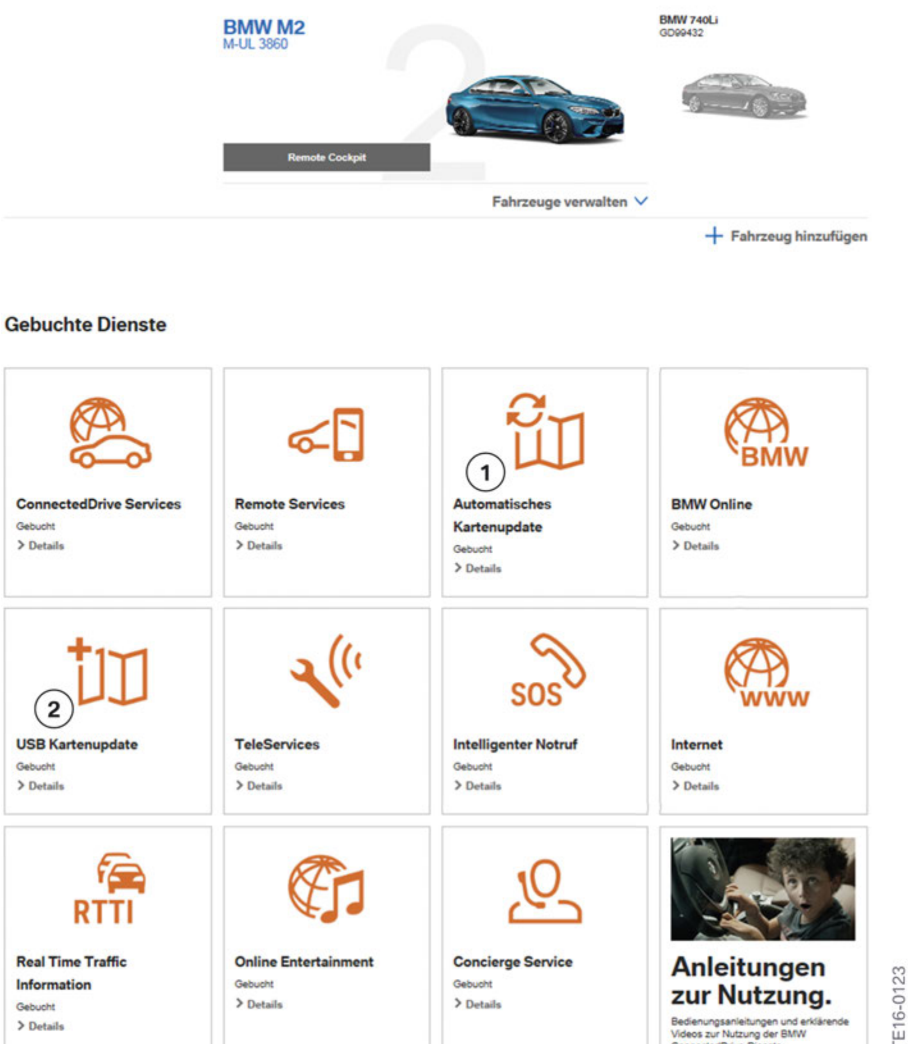
# 2. Navigation Map Data Update

## 2.1. New features of the ConnectedDrive portal for customers

As previously announced in the G12 Navigation System training manual in 2015, the download and setting area for map updates for customers was launched in early 2016.

The ConnectedDrive Store map update is subdivided into two areas. One area is for an automatic map update and a second area called the USB Map Update. Both can be used to download the map data for a full update. Services that have already been booked and that are available can be viewed in the ConnectedDrive portal under:

**Booked services**




Map update area in the ConnectedDrive Store

| Index | Explanation |
|---|---|
| 1 | Automatic map update (booking details) |
| 2 | USB map update (booking details) |

# Infotainment 2016
# 2. Navigation Map Data Update

### 2.1.1. Automatic map update

The **Region setting** feature for the automatic map update was moved to the **ConnectedDrive portal of the customer** when the map update area was launched.

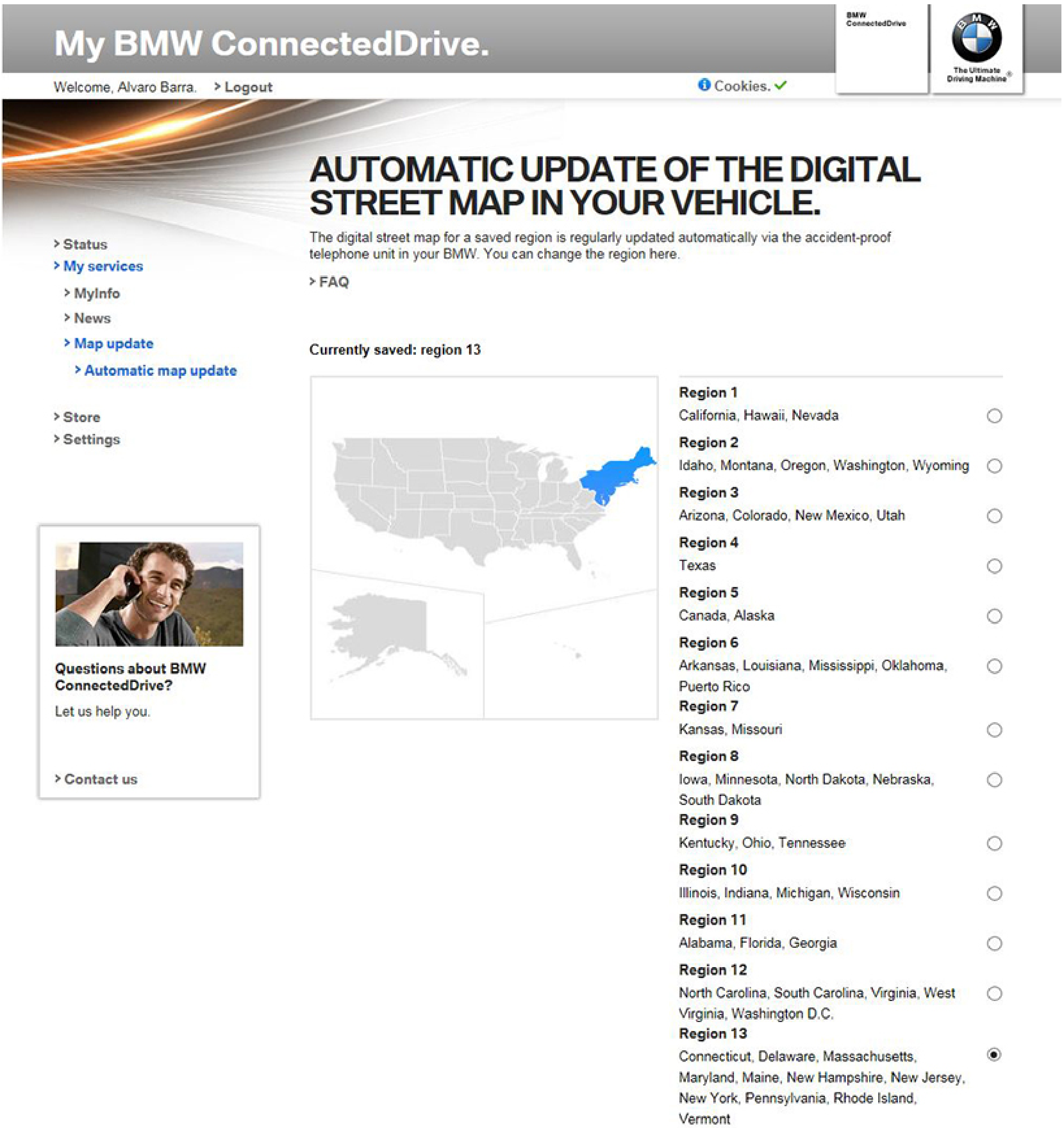
BMW Online made this page available to the vehicle in a dynamic manner. This means that the region setting disappeared when the map update was started in the ConnectedDrive portal from the vehicle, similar to a changed website.

An example of the available (13) map region selections for the US market in the ConnectedDrive portal is displayed on the next page:

# Infotainment 2016

## 2. Navigation Map Data Update

1. If the ConnectedDrive portal is currently displayed with the former corporate identity (CI):

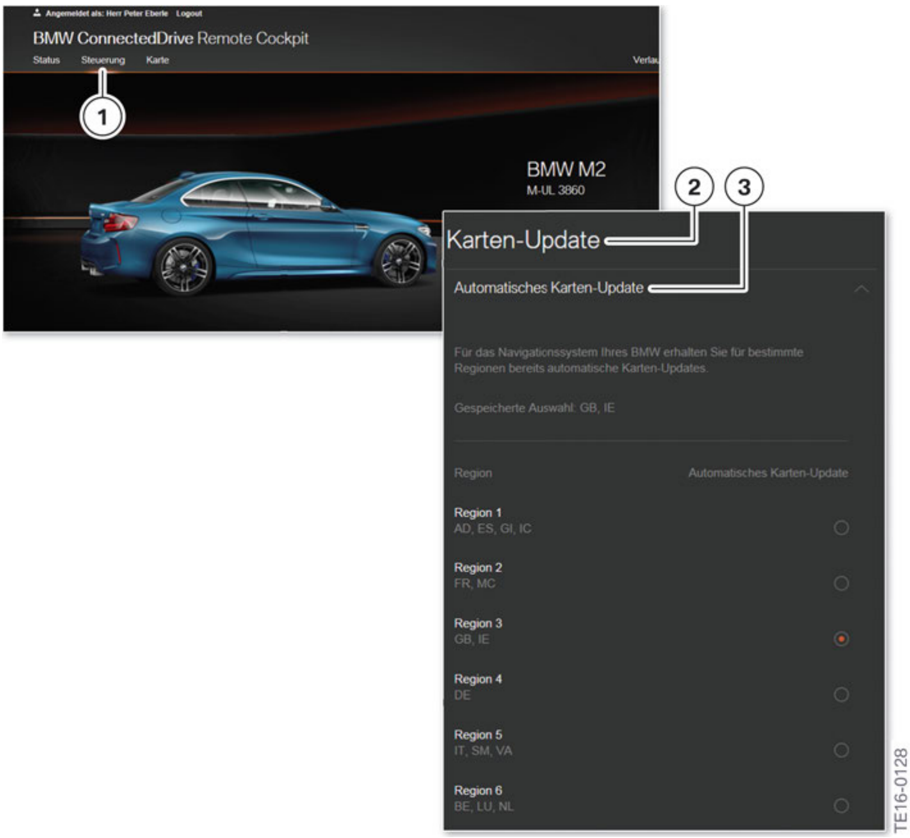



Display of former CI

# Infotainment 2016

## 2. Navigation Map Data Update

2. If the ConnectedDrive portal is already being displayed with the **new CI**:




ConnectedDrive portal with new CI

| Index | Explanation |
|---|---|
| 1 | Control menu item |
| 2 | Map update submenu |
| 3 | Automatic map update with region setting |

These settings, up to now could be made directly via the vehicle iDrive in vehicles equipped with the HU-H2 in 2014 and 2015. Since the changeover, they can **no longer** be called up in the Options area of the vehicle navigation.

In addition to the automatic map update, a **full navigation update** may be necessary to change the region setting.

This is the case whenever the map data has been updated automatically. In this case, a region setting via the automatic map update cannot be fully executed. The data volume would be too large "over the air". A notice requesting a full navigation update then appears in the ConnectedDrive portal and in the vehicle.

The full navigation update can take place via the ConnectedDrive portal or in the service workshop via ISTA-P (programming), or an update can be performed using a USB stick with the map data from the Copy Console (HDD update in ISPI Next).

# 2. Navigation Map Data Update

### 2.1.2. USB map update

The USB map update was already announced to the customer in the price list of the BMW vehicle. Customers can download the map data from the ConnectedDrive Store free of licensing costs for three years.

The customer can move the file, temporarily stored on the PC, to a USB stick with sufficient memory capacity and save it in the vehicle via the USB interface in the center console. (See list of the various Headunit variants in the graphic in section 2.2)

It is no longer necessary to enter the FSC enabling code since the FSC is valid for three years. For security, the enabling code with **20 characters** in the portal is displayed again during the download process and sent to the customer as a text message.

This applies to all vehicles up to Headunit High 2 with ID4. With Headunit High 2, i.e. the Navigation Professional of the latest generation, the code was extended to **128 characters.** Because this would be too long for manual entry, it is stored on the USB stick, **together with the map material** for the respective vehicle.

The map data are downloaded via a special **update manager**. This small program is installed on the PC and can then be used to download the map data and store them on the PC. After the data is transferred to the USB stick, the vehicle data can be written to the vehicle as described above.

# 2. Navigation Map Data Update

The following is a short overview of the **download procedure** in the ConnectedDrive portal:


Update Manager in the ConnectedDrive portal (new CI only)

| Index | Explanation |
|---|---|
| 1 | Control menu in the ConnectedDrive portal |
| 2 | "Manual map update" submenu |
| 3 | Selection of the operating system |
| 4 | Download switch for the BMW Update Manager |

| Index | Explanation |
|---|---|
| 5 | Saving the BMW Update Manager on the PC |
| 6 | Desktop link for the customer |
| 7 | Start of the download procedure for the current map data |
| 8 | Progress bar for the download procedure |

⚠

The formatting of the USB stick is significant. **The FAT32 standard formatting** can be read by all Headunit's. (https://en.wikipedia.org/wiki/File_Allocation_Table).

If the USB stick is correctly formatted, has enough memory capacity, and the files are fully transferred to the USB stick with the right directory and file structure, nothing will stand in the way of a successful update **by the customer**.

## 2.2. Map data update at the service workshop

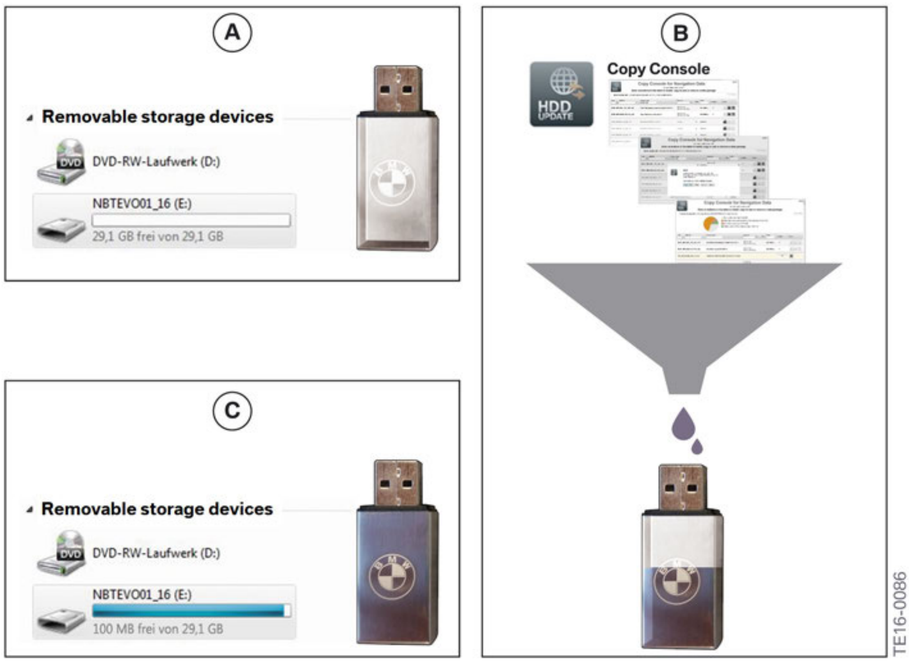
### 2.2.1. USB sticks and memory capacity

There are **3 possibilities** of performing a full map data update of the Headunit in the service workshop: via the USB stick, which the customer obtained through cash sales; via programming; and via a USB stick the workshop creates using Copy Console. The method using **Copy Console** is described here.

To be able to respond quickly to customer wishes for a map update in the service workshop, the workshop should always have a set of USB sticks **(currently 5 sticks)** available with the currently valid map data. The USB sticks could be loaded with the necessary data overnight by the ISPI Admin, for example.

⚠

It is important to note the size of the USB stick (actual available memory capacity). The actual size may vary from the size printed on the USB stick by as much as a "real" 3.1 GB storage capacity. In preparation, the USB stick should be formatted to avoid any resources being wasted by unnecessary files and to have the full memory volume available.




Loading data onto the USB stick using Copy Console

### 2.2.2. Headunit and map data variants

After the latest map data are selected for the respective Headunit, the data can be downloaded and stored on the stick via the Copy Console in the HDD-Update tool (available on ISSS and ISID).

| Model range | Development code | Map data |
|---|---|---|
| CIC | CIC High | PREMIUM Road Map |
| CIC Basic 1 | CIC MID | MOTION Road Map |
| CIC Basic 2 | CHAMP 2 | MOVE Road Map |
| HU-B with Nav | Entry | ROUTE Road Map |
| HU-H | NBT | NEXT Road Map |
| HU-H2 | NBT Evo | EVO Road Map |
| HU-B2 with Nav (2017) | Entry Evo (2017) | WAY Road Map |

Using this method, all current map data can be stored on the USB stick, taking into consideration the respective storage volume of the individual USB sticks.

Below is a comparison of the individual full update variants with the memory capacity requirements for the respective map data on the USB stick.

# 2. Navigation Map Data Update


USB sticks with the individual map variants, which are uploaded via ISSS/ISID and the Copy Console

## 2.2.3. Frequently asked question regarding the Navigation Map Update:

- **Why do I only receive updates for one Region?**
  Currently, Automatic Map Update for one Region is included for free. For information on how to update the complete Navigation Map, visit your local BMW center.
- **How do I know that my BMW is receiving Automatic Map Updates?**
  You will see Automatic Map Update under "Subscribed services" in the ConnectedDrive Store.
- **What should I do when my Automatic Map Update subscription expires?**
  You can renew the Automatic Map Update service in the ConnectedDrive Store.
- **Can I change the region for which I will receive Automatic Map Updates?**
  Yes, you can change your region in the menu under "My services > Map update". Simply select a new region and subsequently confirm the selection. Please keep in mind, that after the region change it may be necessary to perform an full Map Update with the latest available map in order to re-initiate the Automatic Map Update for the new region. Please visit your local BMW dealer for more information.

- **How long does it take to receive an Automatic Map Update?**
  The time varies depending on your BMW's data connection and signal strength and the amount of new information included in the newest map update.
- **How often are Automatic Map Updates released?**
  Up to 4 times per year (quarterly).
- **How do I activate Automatic Map Update?**
  This service is automatically activated for you. The Map Updates will also be sent automatically.
- **Can I receive Automatic Map Updates for more than one region?**
  Currently, it is only possible to receive Automatic Map Updates for one region. If you would like to confirm for which region you are currently receiving map updates or change the region, go to the menu "My services > Map update".
- **How do I know for which region my BMW is receiving Automatic Map Updates?**
  You can view the selected region in the menu under "My services > Map update".
- **Can I continue to use my Navigation System during a Map Update?**
  Of course, the Navigation System remains fully functional during a Map Update.

### 2.2.4. Headunit High special case (CIC)

The data for the CIC Headunit High currently **cannot** yet be loaded using Copy Console. The data format for this special Headunit is **unsuitable** for Copy Console.

⚠

A solution is currently being developed by BMW and will be rolled out in autumn of 2016 with the 2017-1 map data.

As a remedy until autumn, the data of the original BMW USB stick (purchased USB stick) **from cash sales** of BMW Parts Service can be copied onto the workshop USB stick for CIC High.

In this way, the workshop, too, will have a USB stick version of the latest PREMIUM map to use at any time.

Caution! Remember that the update for the **CIC High** Professional Navigation is performed via the vehicle glove box.

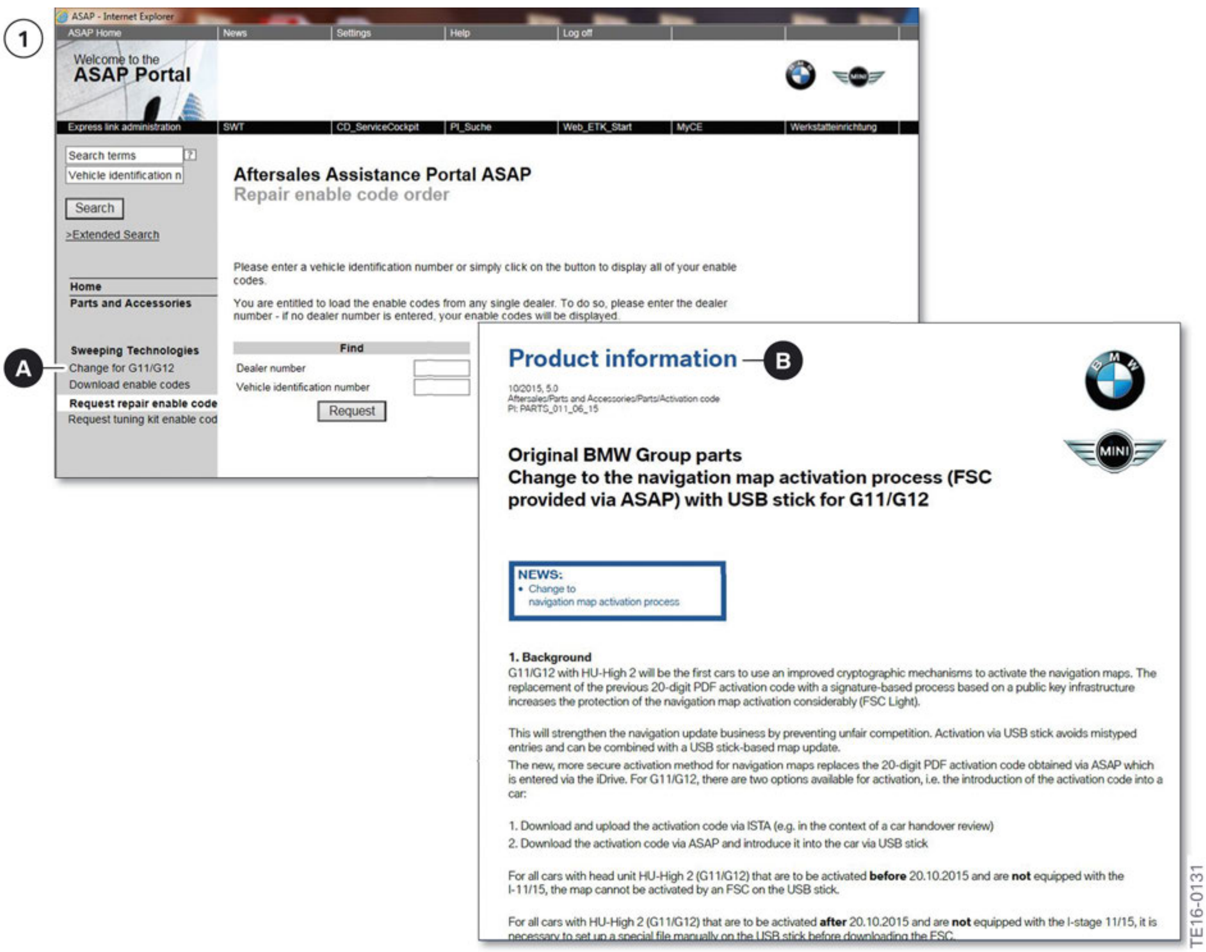
In addition, the PREMIUM DVD variants from the **media package** can be used. The update is then performed via the DVD drive of CIC High.

# 2. Navigation Map Data Update

### 2.2.5. 128-place enabling code FSC

With the G12, the FSC was for the first time increased from 20 places to a total of **128 places**. In HU-H2/ID5 vehicles, the code is now uploaded directly together with the map data and the USB stick or alternatively with the programming system. It is no longer necessary to manually enter the enabling code via the iDrive controller.

Details can be found in the product information on the Sweeping Technology/FSC page in the ASAP portal, where the repair FSC can also be found.




Product information on FSC changes in the ASAP Portal

| Index | Explanation |
|---|---|
| 1 | Sweeping Technologies ASAP Portal area |
| A | Information on the changes to the FSC introduced in G12 |
| B | PI information on changes in G12 with details |

# Infotainment 2016
# 3. Apple CarPlay

Beginning in late summer 2016, customers will be able to use **the iDrive system** to switch to an **Apple user interface** in the vehicle as an alternative to the BMW user interface. A wide variety of features can be opened and on the Apple smartphone that is coupled with the vehicle. (See the 2016 August Service Round Table segment for more information.

A **detailed list** of the individual market launch dates of Apple CarPlay in the various vehicles can be found in section 1.2 “Variants with the new user interface”.

## 3.1. New Apple CarPlay preparation optional equipment

### 3.1.1. ConnectedDrive markets

The ConnectedDrive customer can order the **Apple CarPlay Preparation (SA 6CP)** optional equipment when the "ConnectedDrive Service" (SA 6AK) and "Navigation Professional System" (SA 609) optional equipment is present in the vehicle. This optional equipment is called a **preparation** because customers require **their own iPhones** for implementation of the feature in the vehicle.

Later booking via the customer ConnectedDrive portal or via the ConnectedDrive store in the vehicle is currently being tested.

### 3.1.2. Headunit's for Apple CarPlay

The "Apple CarPlay Preparation" (SA 6CP) optional equipment will initially be offered **exclusively** in combination with the HU-H2 Navigation Professional (SA 609) Headunit's. This will be expanded as of 03/17 with the CarPlay offering in connection with the Navigation Business (SA 606/SA 6UN) **HU-B2** Headunit and Navigation Plus (SA 6UP) for F48.

An additional offering for the Radio Professional HU-B2 (Entry Evo Media) is currently being tested.

# Infotainment 2016
# 3. Apple CarPlay

## 3.2. Prerequisites for the iPhone for CarPlay

To ensure smooth connection setup, an Apple iPhone of version **iPhone 5** or later is required. The operating system required for a smooth connection setup is at least **iOS 9.3**. BMW recommends the version iOS 9.3.3, which was be available from Apple in July 2016.

Other Apple CarPlay devices such as iPad®or iPod® cannot be connected with the vehicle via Apple CarPlay.

The compatible devices can be looked up on the Apple CarPlay page regarding the CarPlay topic:

**http://www.apple.com/ios/CarPlay**


Recommended Apple CarPlay iPhones for connection setup with Apple CarPlay

## 3.3. Connection setup

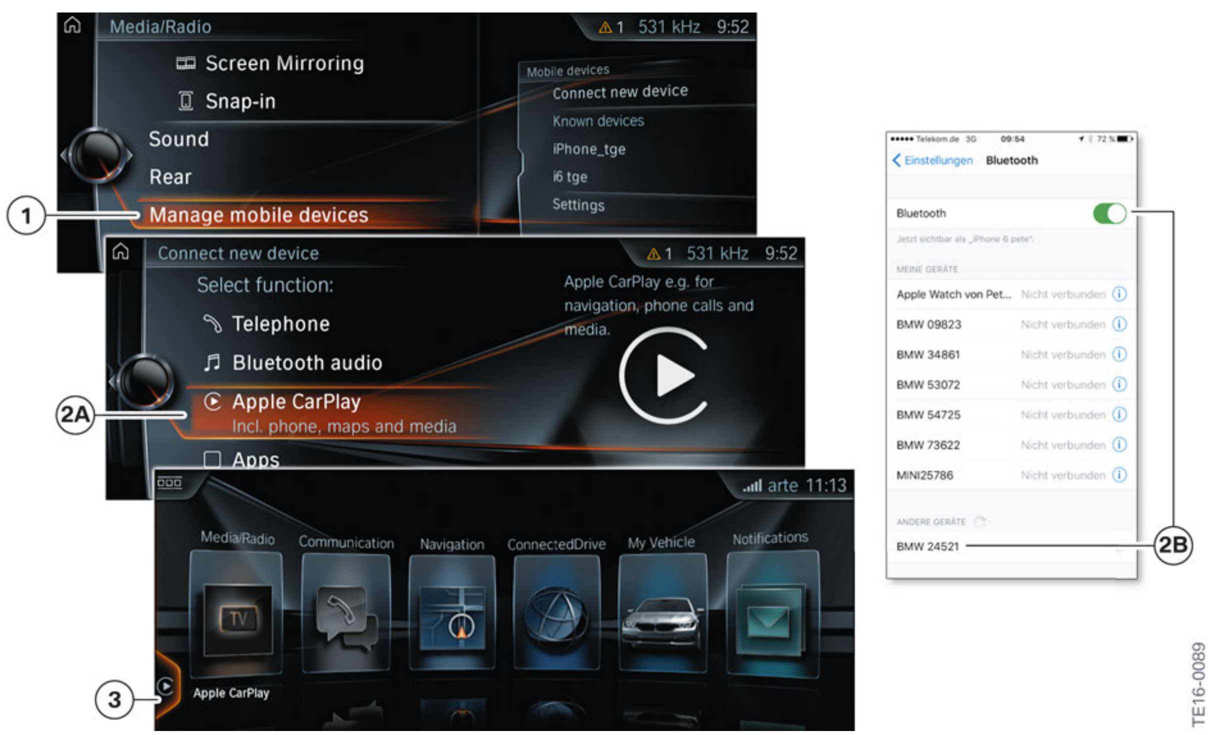
### 3.3.1. Connection manager

The connection to Apple CarPlay is set up similarly to setting up a Bluetooth connection with a cell phone. The customer merely selects **Apple CarPlay** instead of the "Telephone" function.

A WLAN connection is established in the background, invisible to the customer. This means that, if present, the Internet hotspot in the vehicle (via the TCB2) **cannot** be used since the iPhone automatically connects to CarPlay and the corresponding WLAN function will already be in use.

CarPlay would have to be **manually** deactivated by the customer on the iPhone in order to be able to surf on the Internet via the vehicle hotspot.

The WLAN Direct connection to the Screencast/Mirrorcast features with other mobile devices (Android) are possible in the vehicle even without manual deactivation.




Changeover from BMW to the Apple CarPlay user interface

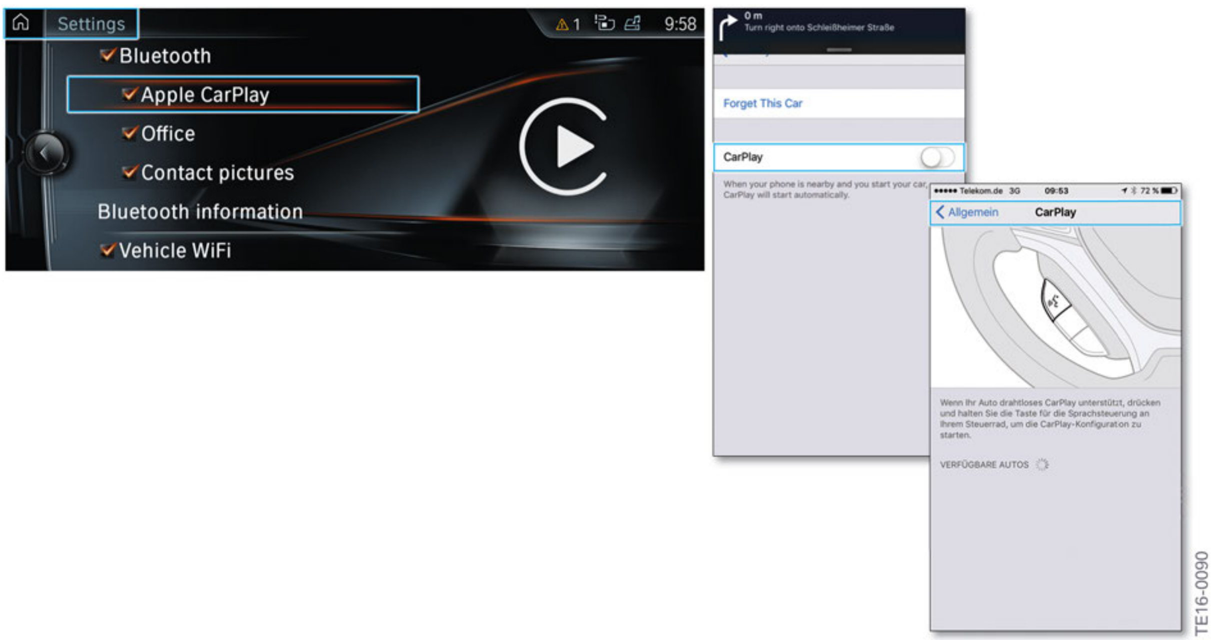
| Index | Explanation |
|---|---|
| 1 | Media/Radio menu: Add new device |
| 2A | Apple CarPlay (connection manager selection "Add new device") |
| 2B | iPhone Bluetooth connection setup with the vehicle; Bluetooth "active" |
| 3 | Successful connection setup visible in the main menu of the CID Apple CarPlay |

# 3. Apple CarPlay

### 3.3.2. Settings

In the "Settings" menu of the vehicle, the Bluetooth connections for "Office" and "Contact pictures" are shown in addition to the active "Apple CarPlay" connections.

On the iPhone, the active CarPlay connection is shown in the "Settings" - "Bluetooth" menu (see figure in section 3.3.1) and in the "General" – "CarPlay" menu.




Apple CarPlay settings in the vehicle and on the iPhone (manual CarPlay switch in the iPhone)

### 3.4. Activating Apple CarPlay

After CarPlay is activated, the customer can directly select the "Apple CarPlay" menu item (CarPlay symbol) on the left in the main menu by turning and pressing the iDrive controller.

In addition, Apple CarPlay can be jumped to directly from the following iDrive menus:

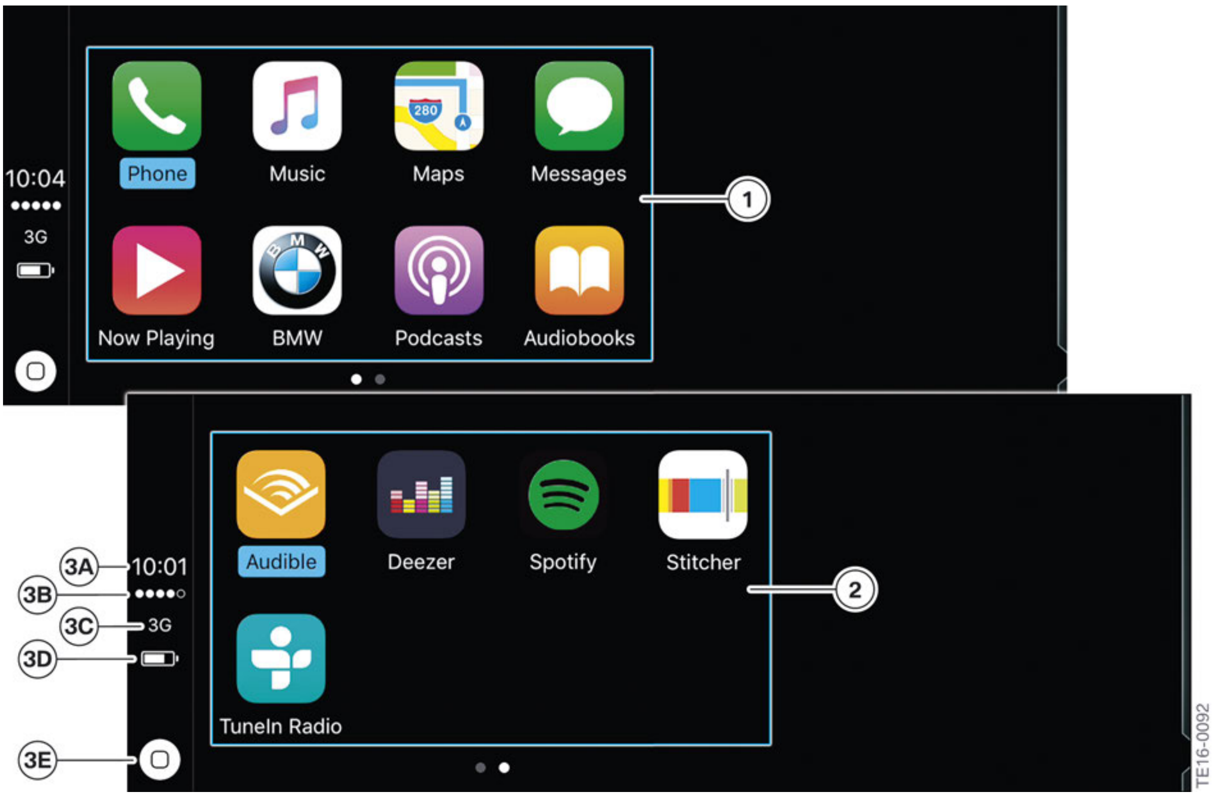
- Media/Radio
- Communication
- Navigation menu


Direct jump from the BMW navigation menu to the Apple CarPlay user interface

| Index | Explanation |
|---|---|
| 1 | Apple CarPlay is **active** in the BMW iDrive main menu |
| 2A | Example: Opening Apple CarPlay via the navigation menu of the iDrive system |
| 2B | Direct jump example: Navigation "Destination input – Maps in Apple CarPlay" submenu |

## 3.5. Apple CarPlay menu structure

Overview of the two main menu pages of the Apple CarPlay user interface:



Comparison of the two main menu pages in Apple CarPlay

| Index | Explanation |
|---|---|
| 1 | Apple CarPlay main menu |
| 2 | Expansion of the Apple CarPlay main menu for third-party apps |
| 3A | Time display |
| 3B | Display of field strength of the Apple device connected via Bluetooth |
| 3C | Display of currently possible upload and download link speed, smartphone |
| 3D | Status display of the state of charge of the coupled Apple iPhone |
| 3E | Home button for returning to the Apple CarPlay main menu |

# Infotainment 2016

# 3. Apple CarPlay

## 3.6. Telephone and messages

Telephone calls and the reception and writing/dictating (via SIRI) of messages are fixed features of Apple CarPlay. Apple customers who are already familiar with the iPhone will find these features easy to operate.

The symbols are entirely aligned with the iPhone/Apple philosophy. Favorites, recent calls, the address book, etc., are intuitive to find.

### 3.6.1. Telephone system

In addition to calls via standard, digital telephony, calls can also be placed in excellent quality from Apple device to Apple device via the Apple-own **FaceTime Audio** call feature.


Telephone menu of Apple CarPlay

| Index | |
|---|---|
| 1 | Explanation of the telephone symbol in the Apple CarPlay main menu |
| 2 | Submenu |
| 3 | Contact selection |
| 4 | Selection of "recently used contacts" |
| 5 | Current phone call |

### 3.6.2. News

Messages can be opened and written in a variety of ways. The SIRI voice control feature supports the writing of text messages via speech-to-text voice input, for example.

### 3.6.3. Main and additional telephones

In the Apple CarPlay menu, it is also possible to use main and additional telephones without any problems. In the following example, an Android device was defined as the additional telephone and the iPhone as the main telephone or as the CarPlay device.

The behavior of each of these devices coupled to the vehicle when Apple CarPlay is activated in the vehicle can be seen in the following screen sequence:


Incoming telephone calls (additional and main telephones in Apple CarPlay)

| Index | Explanation |
|---|---|
| 1 | Apple CarPlay menu is active (Navigation in this case) |
| **Additional telephone** | |
| 2 | Incoming telephone call, CID changeover to a pop-up window for an incoming call (additional telephone) |
| 2B | Coupled additional telephone in the "Mobile devices" menu |
| **Main telephone** | |
| 3 | Incoming telephone call, display in the CID of the main telephone coupled to Apple CarPlay |
| 3B | Coupled main telephone (iPhone) in the "Mobile devices" menu |

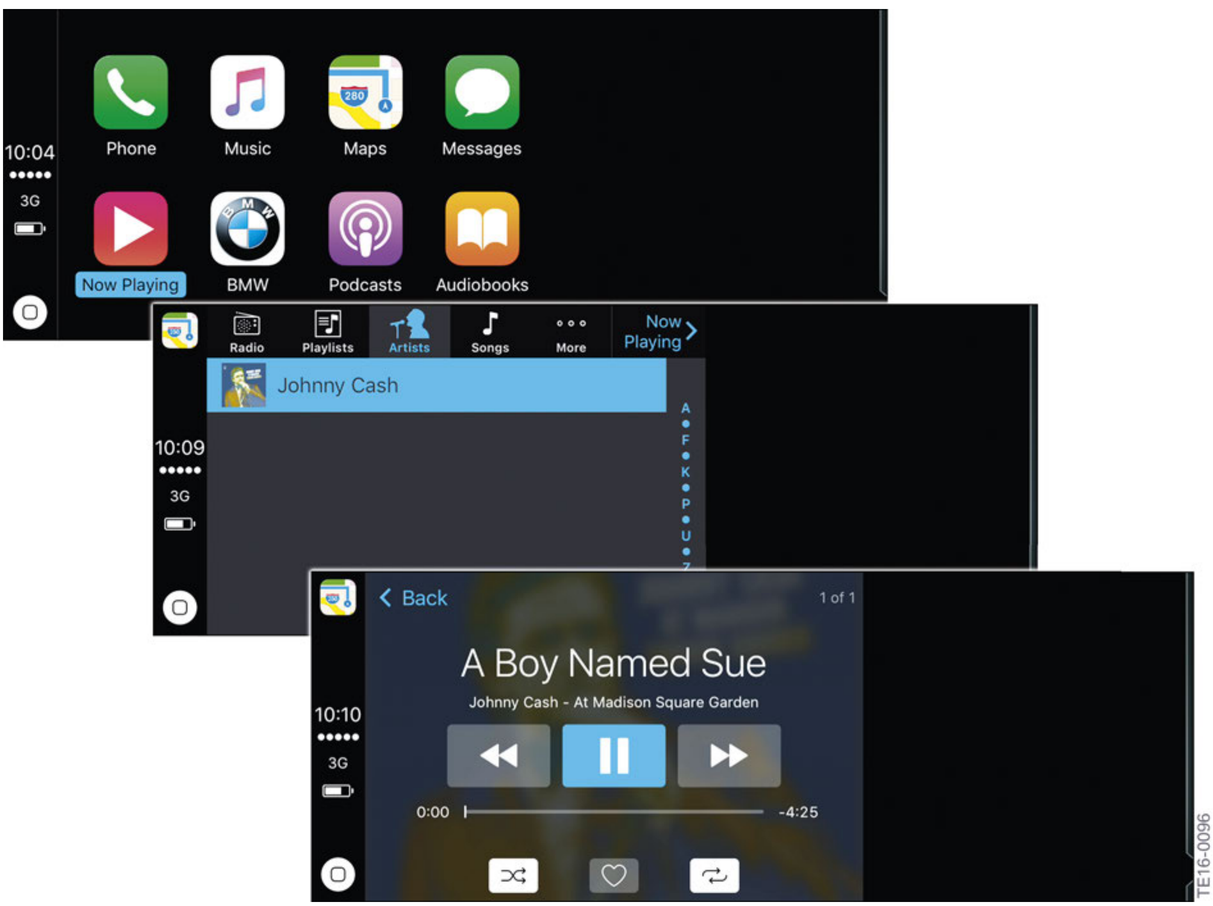
## 3.7. Music player and more

### 3.7.1. Now playing

In the "Now Playing" menu item, playback of the music player in the iPhone is actively displayed. The source on the smartphone is of secondary importance. Regardless of whether browser or YouTube content is currently being played on the iPhone, the content is now available in the vehicle.

In addition, "Now Playing" also plays the music stored on the iPhone (link to Music menu item) and audio playback from streaming services.

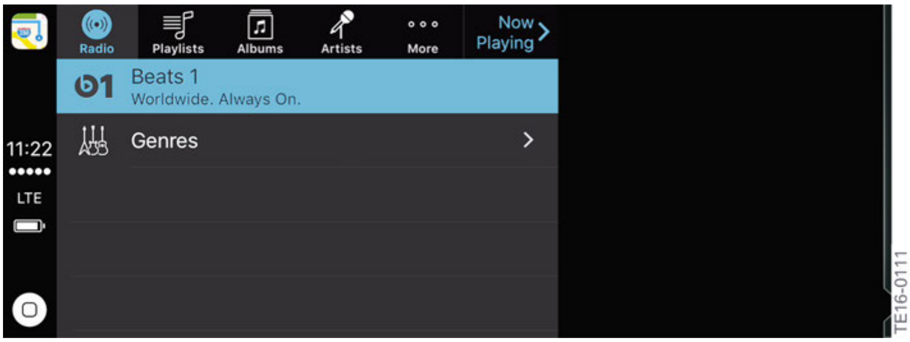
This also applies to third-party providers and streaming services, such as Napster, that currently are not supported directly by Apple CarPlay. However, the functionality is restricted in these cases and not as diverse as for implemented third-party apps (see section 3.7.3).



Activated Apple Carplay music player

### 3.7.2. Radio

In the "Radio" submenu, free radio stations or stations booked via Apple Music are displayed and can be activated here. Radio stations already booked from Apple Music can also be played back here. However It is not possible to purchase Apple Music through here.




Web radio via the iPhone displayed in Apple CarPlay

In the "Music" menu item, music from iTunes is played that either is stored on the device or is stored in the iCloud and then streamed to the vehicle.

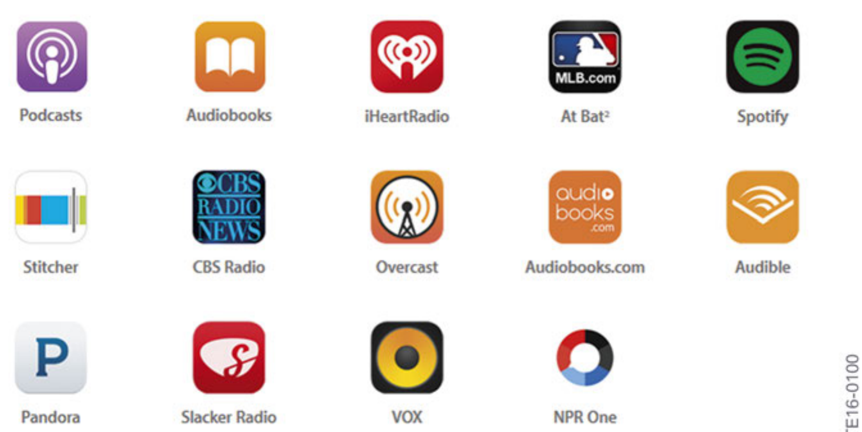
In addition to music playback with "Now Playing" and "Music", the Podcasts and Audiobooks apps are supported as well.

### 3.7.3. Third-party offering in CarPlay

In addition, third-party apps are supported by Apple CarPlay. However, unlike "Apps" (SA 6NR) or A4A (Application for automotive), the full range of third-party apps on the market is not supported.

A short list of some of the apps that can be used in CarPlay is provided by Apple under: http://www.apple.com/ios/carplay.

The following list shows the third-party providers apps supported by Apple CarPlay according to Apple.




Apps supported by Apple CarPlay according to the CarPlay homepage

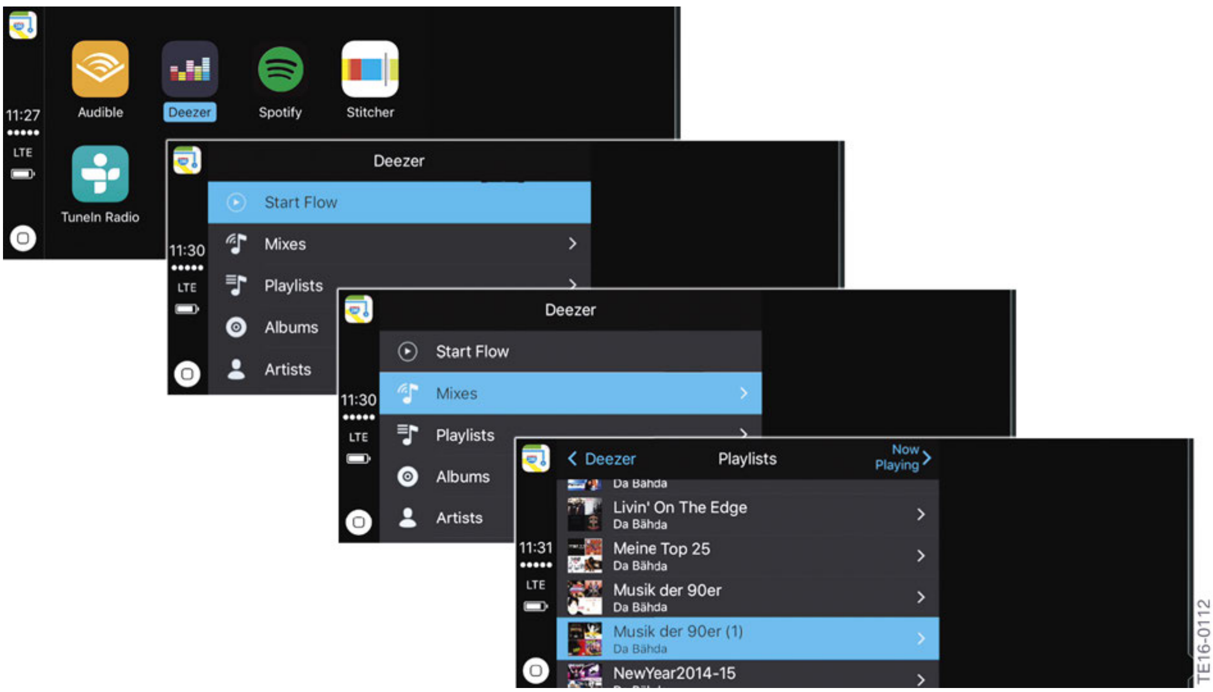
# Infotainment 2016

# 3. Apple CarPlay

A comparison with the main menu in section 4.4. shows that **other** third-party apps are supported in part as well. A full list of all apps supported by CarPlay is not available at this time.

An example of third-party apps is the offering from **Deezer**, here actively displayed in the vehicle by Apple CarPlay. Examples of individual Deezer sub-functions are shown here.

The display shown here may vary from the actual display because apps undergo continuous change as they are developed further or adjusted to the iOS.




Active Deezer app in Apple CarPlay

## 3.8. Navigation

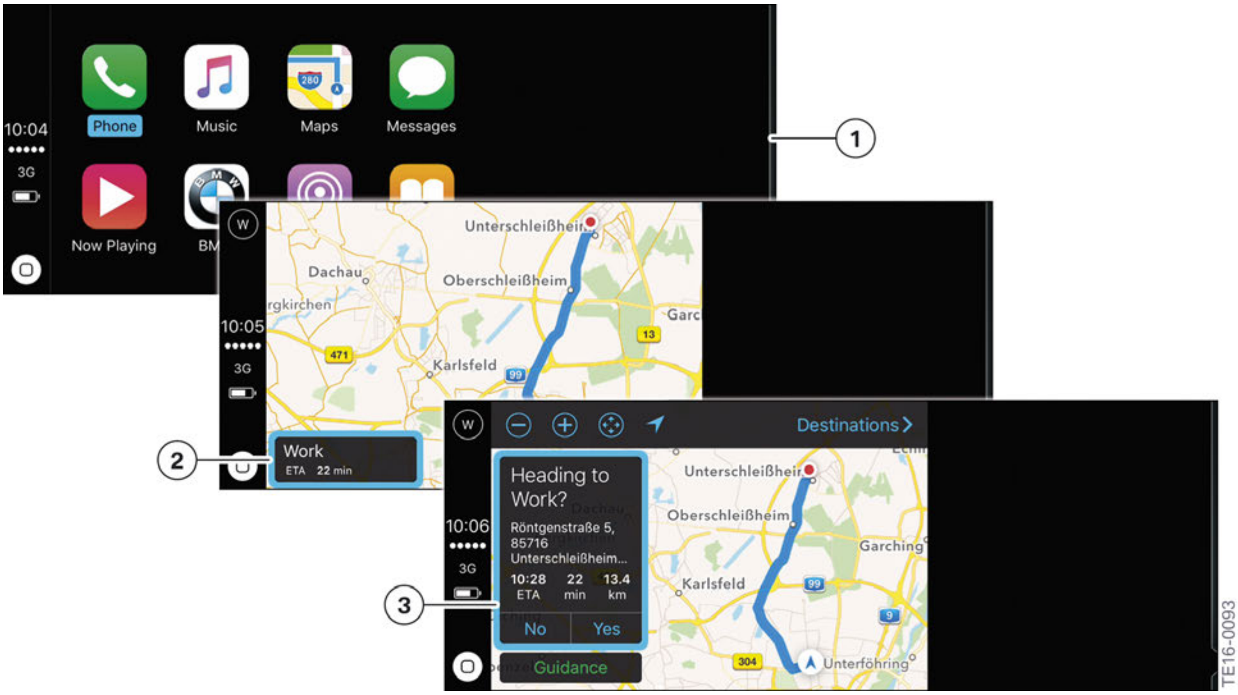
### 3.8.1. Opening the navigation menu

In addition to the telephone and music player functions, Apple CarPlay also features an online navigation system and an interface to the BMW Navigation feature.

For the online navigation, the SIM card contained in the iPhone is used with the data volume of the customer. For this type of navigation, full network coverage is recommended since there is no offline buffer memory.

After activation, the learning navigation of Apple CarPlay automatically recognizes the next segment of the trip and suggests it. The "Heading to work" example is shown below.

If this is the desired navigation destination, it merely needs to be confirmed. Alternatively, the destination suggestion can be declined and a new destination can be selected using the entry assistant, interactive map or SIRI voice control.

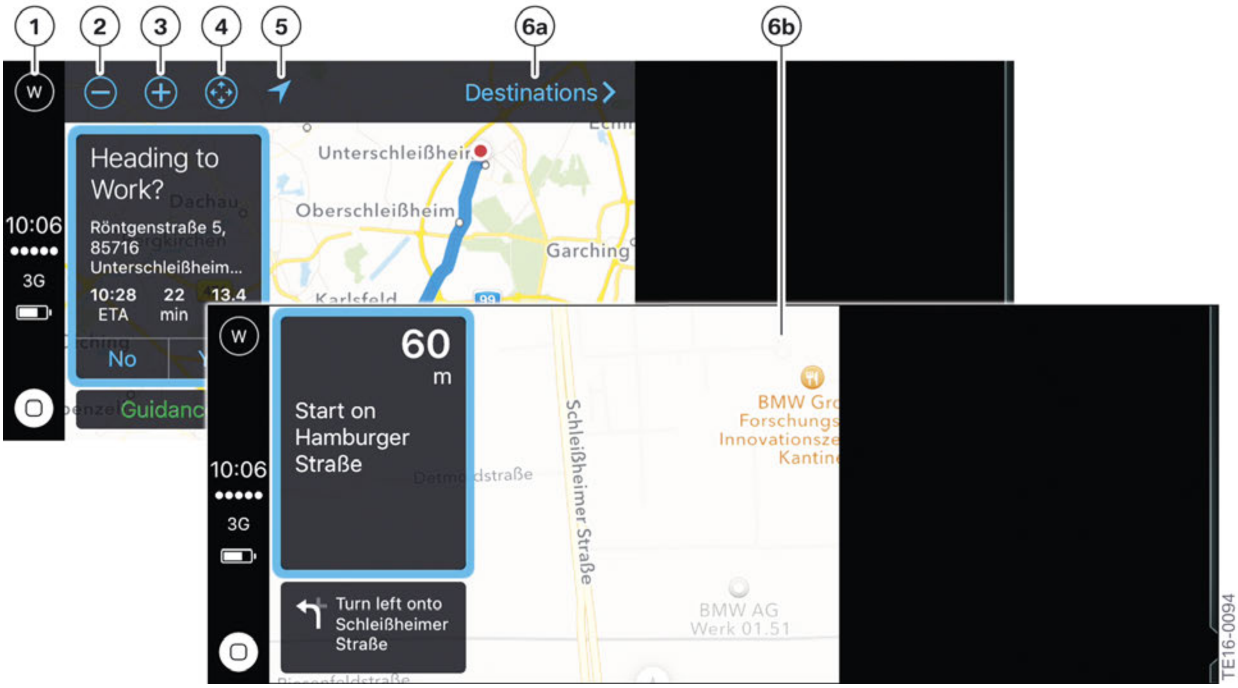



Destination suggestion by the learning Apple CarPlay Navigation

| Index | Explanation |
|---|---|
| 1 | Main menu |
| 2 | Destination suggestion after activation of the online navigation in Apple CarPlay |
| 3 | Accepted destination suggestion in Apple CarPlay |

### 3.8.2. Symbols and menus

The navigation menu contains a large number of additional symbols for better orientation. Some of these symbols are static or are adjusted automatically; others can be selected via iDrive or touch.



Navigation menu with special Apple CarPlay symbols

| Index | Explanation |
|---|---|
| 1 | Compass |
| 2 | Enlarge map scale (zoom out) |
| 3 | Reduce map scale (zoom in) |
| 4 | Interactive map |
| 5 | Northward pointing arrow |
| 6a | Last destination |
| 6b | Not included with launch (future interface to BMW Navigation, called "POI-handover", is currently being tested for 03/2017 measures) |

# Infotainment 2016

## 3. Apple CarPlay

### 3.8.3. Guiding and learning navigation

As is common in other navigation systems, Apple CarPlay Navigation, too, shows additional guiding information in the form of an arrow display in an assistance window. These accompany the customer all the way to the final destination.

In addition, suggestions from the Apple learning navigation are regularly made in the field above the guiding information. Suggestions that affect the time and place are generated automatically by the system (see figure in section 3.9.2 under the symbol bar).

Navigation menu in Apple CarPlay with guiding information

## 3.9. Operation

### 3.9.1. Controller and touch

All iDrive controllers currently offered by BMW

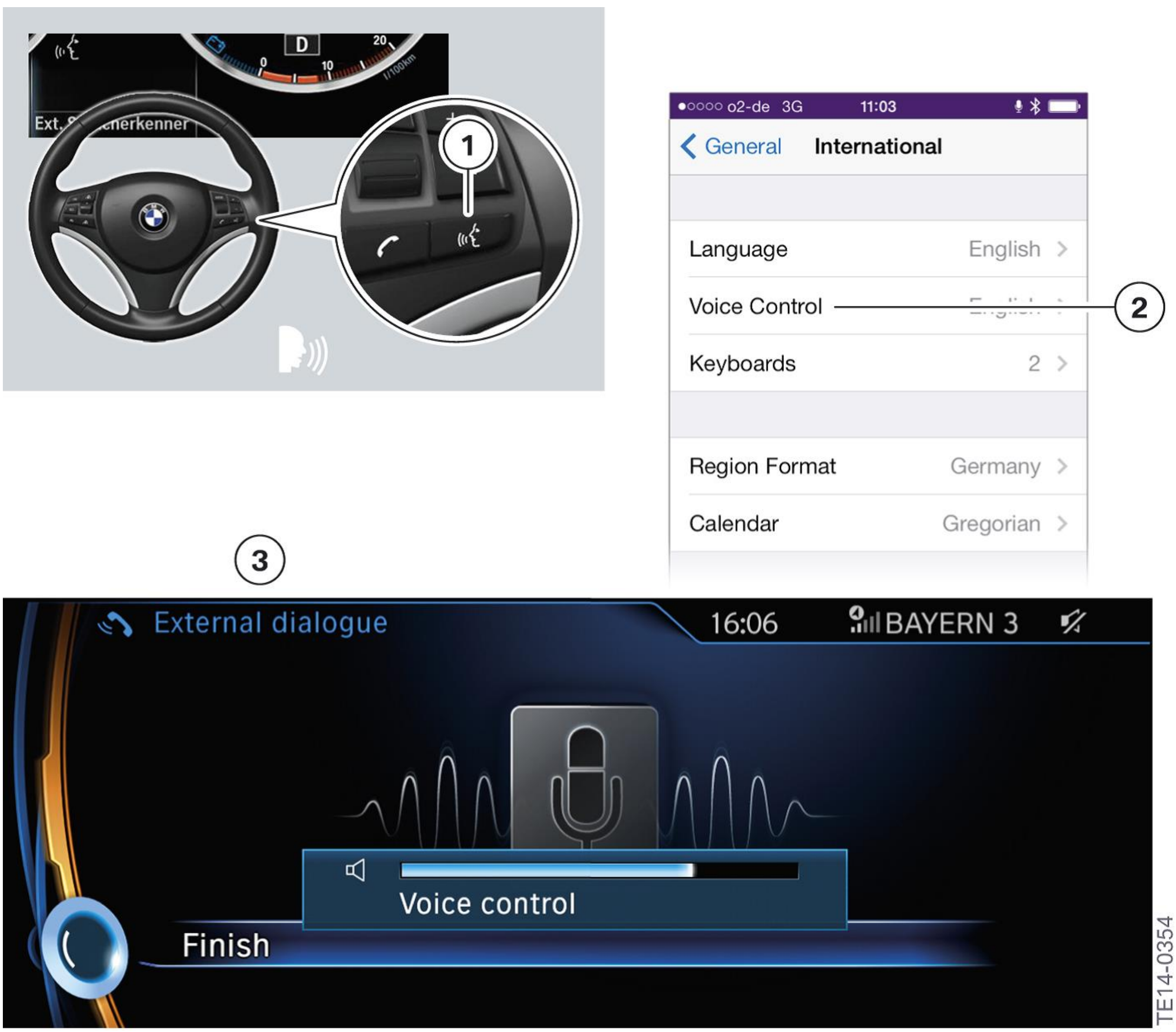
Apple CarPlay can be operated in the vehicle via the iDrive controller and (if present in the vehicle) also via the Touch Control Display CID.

### 3.9.2. Voice control

In addition to the controller and Touch CID, voice control also works in CarPlay. In this case, however, the BMW voice control found on the Headunit's is not used; instead, all functions are operated via Apple's own **SIRI voice control**, as already familiar from the iPhone.

SIRI has been interacting with BMW vehicles with the large Headunit – HU-H or Navigation Professional (SA 609) – since 7/13. In addition, this features has also been available in conjunction with HU-B, the Headunit Basic, and with Business Navigation (SA 606) since July 2014.

Until now, however, activation of SIRI in the vehicle was displayed, but not any visual feedback.



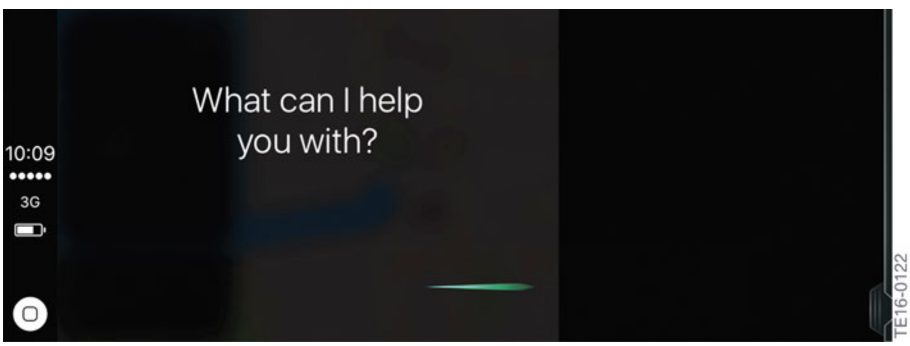

Navigation menu from 2013

| Index | Explanation |
|---|---|
| 1 | "Push to talk" button on the steering wheel (press long for about 2 s) |
| 2 | The "Language" iPhone setting menu determines the SIRI language. |
| 3 | Voice dialogue of Apple CarPlay with a previously connected iPhone |

### 3.9.3. New SIRI features

For the first time, Apple CarPlay not only displays the active SIRI voice control feature but also shows it's contents.

Questions are displayed on the CID, corrections to text messages, for example, are visualized and changes between functions, such as from navigation to music, are now visible to customers.



Active voice input via Apple SIRI

### 3.10. Changeover between user interfaces

To be able to use the "Apps" (SA 6NR) feature as before, a manual changeover to Apps is required in the "Mobile devices" menu. The Apple CarPlay function is deactivated for this and must be reactivated later when it is needed.


Customer alternatives: CarPlay or BMW iDrive apps

| Index | Explanation |
|---|---|
| 1A | Apple CarPlay |
| 1B | BMW iDrive (including apps) |

# 4. The New BMW Connected App

## 4.1. Content

The new BMW Connected App contains the contents of other apps that are already familiar and replaces these with a single app. It will replace the following apps gradually over the course of 2016.

- BMW Remote App
- BMWi Remote App
- BMW Connected Classic (BMW Connected App was renamed)
- BMW Remote App (G12)

Thus, the new app contains the already familiar function blocks and **new** elements.

**New elements** are the learning navigation including the trip planner and the assistant function with a reminder windows that interact with the vehicle navigation.

In addition, the BMW Connected App on the iPhone makes it possible to access destinations in Google Maps or Foodguide, for example, via "**Parts Symbol**" and to then transfer these to the vehicle.

# Infotainment 2016

# 4. The New BMW Connected App

## 4.2. Schedule

### 4.2.1. Launch Scenario

The new BMW Connected App was already launched in the US market in April 2016.

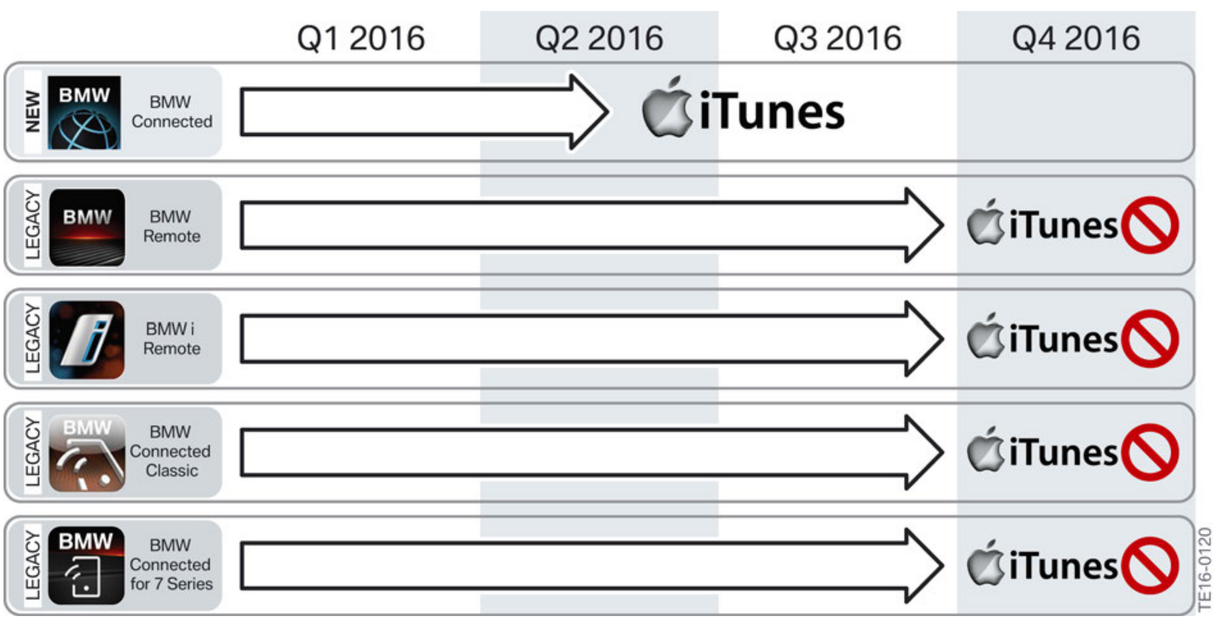
### 4.2.2. Discontinuation of previous apps

The **two remote apps** (BMW and BMWi Remote App) will be fully integrated in the update versions of the **new Connected App** over the course of the year.

Initially, only sub-functions of the various remote apps will exist in the new Connected App.

After full integration, the separate versions of the two remote apps will be removed from the iTunes Store. This is expected to occur by late 2016 or early 2017.

The same applies to **BMW Connected Classic**. This app, too, will continue to be offered in the iTunes Store until it is fully integrated in the new app, and will then be removed from the iTunes Store, most likely in early 2017.




Introduction scenario of the new BMW Connected App and the discontinuation of previous apps

### 4.2.3. New Features

For further information please visit the BMW ConnectedDrive page for the US market, which shows detailed screenshots of the app.

www.bmwusa.com/standard/content/innovations/bmwconnecteddrive/connecteddrive.aspx#view_apps

In addition to an overview of the available apps, this page contains additional information and an **FAQ list** showing the most important questions on the BMW Connected App for downloading as a PDF file.

# Infotainment 2016

# 5. IFTTT

## 5.1. Origin and meaning

In January 2016 BMW Labs (labs.bmw.com) released a new Widget based on the IFTTT technology (pronounced “ift” as in “gift” and is an abbreviation for “IF This Then That”) and describes the feature with which different services (such as Facebook, Evernote, Dropbox, etc.) can be connected with simple conditional instructions from the Internet. For this purpose, BMW uses the IFTTT.com service together with it's ConnectedDrive offering.

Highlights

- **What is BMW Labs?**
  “BMW Labs enables you to try out new experimental ConnectedDrive services which we are currently working on. Enjoy shaping the future of mobility with us. Keep in mind about early prototype or ß-versions:” Please note that the products released by BMW labs are experimental and early prototypes.
- **How does this work?**
  With the BMW Widgets and IFTTT integration, customers can setup “recipes” through the IFTTT site or app. For example, a recipe would be “IF my BMW reports itself as parked THEN send me an email with it's parked location”, “IF I reach a certain location THEN send an email to my spouse”.
  It can be used to do some cool things like when you get home your front door automatically unlocks for you and your lights turn on or when I get to work, your ringer shuts off automatically, etc.
- **Where to go for more info?**
  Please visit labs.BMW.com for more information. Especially important is the “Instructions” document about how to get started – this will help to answer many customer questions about how to connect.

Logo of the IFTTT® company, written out

IFTTT is a service provider from San Francisco, USA, that was developed by Linden Tibbets and launched in 2010. The IFTTT servers are hosted on Amazon Web Services. An advantage of IFTTT is that the activated services can be controlled via the IFTTT server. The vehicle merely sends the trigger signal, nothing else.

This means that ConnectedDrive in the vehicle is not opened by IFTTT and is therefore also not susceptible to malware. These expanded services with IFTTT make is possible to interconnect an almost countless number of services. A further advantage is that these programs can be executed automatically. Thus, the user does not even need to hold a smartphone to start an action.

### 5.1.1. Data security

A security feature of IFTTT is that the activated services are controlled via the IFTTT server. The vehicle merely sends the trigger signal, nothing else. This means that Connected Drive in the vehicle is not opened by IFTTT and is therefore also not susceptible to malware.

These expanded services with IFTTT make is possible to interconnect an almost countless number of services. A further advantage is that these programs can be executed automatically. Thus, the user does not even need to hold a smartphone to start an action.

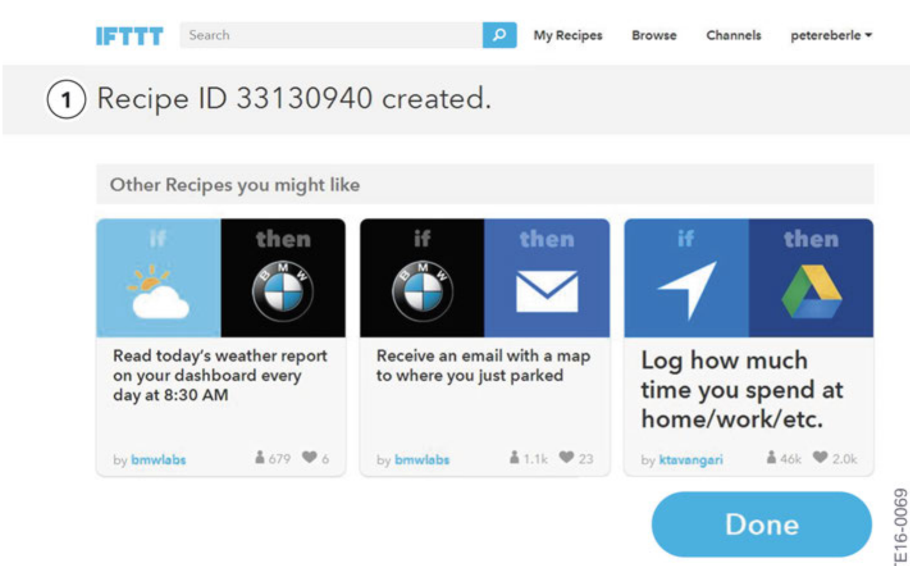
## 5.2. IFTTT at BMW

### 5.2.1. Portal

On January 19th of 2016, the "BMW Labs" portal went online. This gives customers the opportunity to test new services during development and thus actively contribute to the future of BMW ConnectedDrive services. The first ConnectedDrive service being made accessible via BMW Labs is the collaboration with the IFTTT service provider for use of IFTTT in BMW vehicles. IFTTT is a free service via which web applications and intelligent devices from the Internet of Things can be interlinked.

BMW customers can reach the portal via the BMW Labs website: https://labs.bmw.com or for more information and frequently asked questions please visit https://labs.bmw.com/faq.html

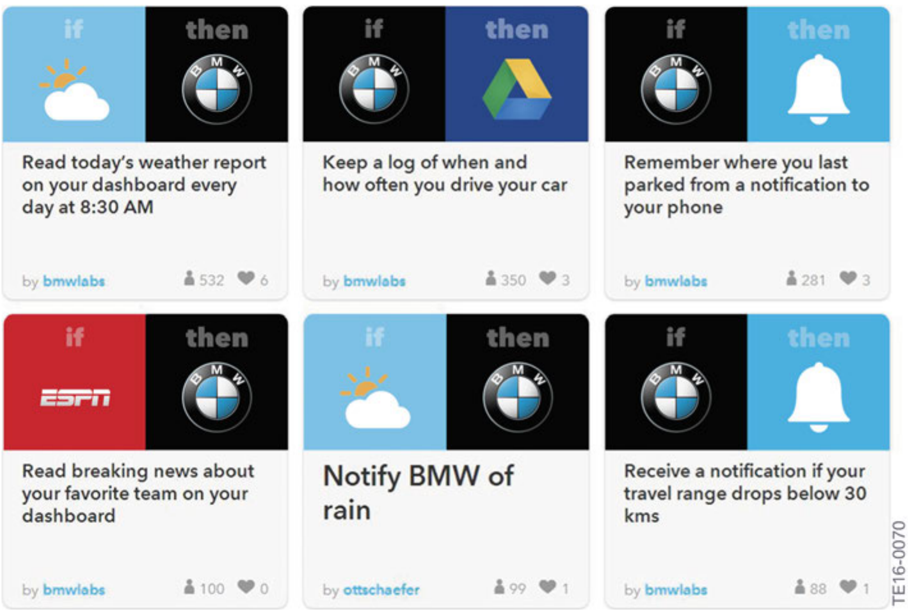
The user can create rules – so-called recipes – in the portal via which the desired triggers can be freely combined with associated actions.




Recipe created in IFTTT

| Index | Explanation |
|---|---|
| 1 | Recipe ID 33130910 was successfully stored in the IFTTT portal |

These triggers are sent to the BMW portal via the vehicle and are then executed or recorded. The user can select from over 260 different services such as Twitter, Facebook, Google Drive, e-mail and text message – as well as intelligent smart home devices such as Hue, Nest or Netatmo. Via the IFTTT recipe, the user's own vehicle can be linked to these services at will.




Excerpt from several recipes = links with the vehicle

For example, the exterior lights of a house can be switched on automatically as the BMW vehicle approaches the house – and in addition, the garage door could be opened and the air conditioning of the house could be turned on.

The family could be informed when the vehicle is in the vicinity of a certain location, such as to announce picking up of children from school. If the vehicle is parked, it's position can be stored as a Google Maps link as a reminder of it's location. A type of log book is also offered in Google Drive for recording individual trips.

### 5.2.2. Equipping the vehicles

Vehicle requirements:

- BMW vehicles built between 07-2013 and 10-2015 (excl. 2015 7 series)
- Vehicle equipped with active ConnectedDrive Services (SA6AK)
- BMW ConnectedDrive Account with mapped vehicle, register/login ConnectedDrive page
- Vehicle equipped with Navigation Professional (SA609: NBT and NBT Evo)
- BMW Labs Widget active in split screen of car’s control display Instructions

The associated BMW Labs widget can be selected in the "BMW Online widgets" menu item of the split screen menu on the BMW Control Display.

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TE16-0073

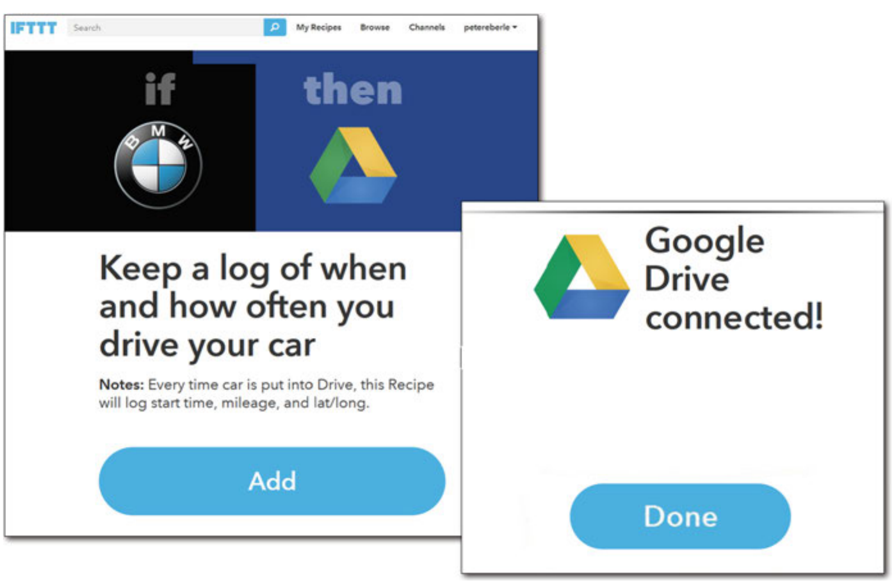
BMW Online widgets as a menu item in the split screen menu

Use of this feature requires the Headunit High 2 in combination with Navigation Professional (SA 609) and the ConnectedDrive Services (SA 6AK) optional equipment including BMW Online as well as previously logging on to IFTTT. **At the moment, only a certain build period is being supported.** Details can be found on the BMW Labs website:

https://labs.bmw.com

To activate the individual recipes, the ConnectedDrive password of the customer and a login into the IFTTT platform is required. Detailed instructions can also be found on the BMW Labs website at https://labs.bmw.com.

The recipes can be selected and activated on the IFTTT platform.

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Selection of the recipe = Google®Drive and it's connection

### 5.2.3. Trigger in the vehicle

In the vehicle, triggers are activated as follows in the BMW Online Widgets menu:


Successful triggering of the desired "recipe" with "Trigger sent"

### 5.2.4. National-market versions

BMW Labs will initially test IFTTT in BMW vehicles with a beta version in English in the USA, England, France, Spain, Germany and Australia. The rollout is currently taking place in Norway. Additional countries are being planned as listed on the website:

https://labs.bmw.com

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